最近在机器学习中应用最优传输方面的进展

Eduardo Fernandes Montesuma, Fred Maurice Ngolè Mboula, and Antoine Souloumiac,

摘要—近年来，最优传输被提议作为一种概率框架，用于机器学习中的概率分布比较和操作。这源于其丰富的历史和理论，并为机器学习中的不同问题提供了新的解决方案，如生成建模和迁移学习。在这篇综述中，我们探讨了2012年至2023年间最优传输在机器学习中的贡献，重点关注四个子领域：监督学习、无监督学习、迁移学习和强化学习。我们进一步强调了计算最优传输及其扩展（如部分最优传输、不平衡最优传输、Gromov最优传输和神经最优传输）的最新发展，并讨论了它们与机器学习实践的相互作用。

索引词—最优传输、Wasserstein距离、Sinkhorn散度、公平性、生成建模、字典学习、聚类、域适应、分布强化学习、贝叶斯强化学习、策略优化

1 引言

O PTIMAL运输是数学领域的一个成熟分支，由加斯帕尔·蒙日[1]和列昂尼德·康托罗维奇[2]的工作奠定基础。自其诞生以来，这一理论对科学做出了重大贡献[3]、[4]、[5]。在这里，我们研究最优运输（OT）如何在机器学习（ML）的不同问题中发挥作用。最优运输在机器学习（OTML）是机器学习社区中一个不断增长的研究领域。事实上，最优运输对机器学习至少通过两个视角是有用的：(i) 作为损失函数，(ii) 用于操作概率分布。

首先，OT 定义了一个 *metric between distributions*，被称为不同的名称，如 Wasserstein 距离、Dudley 度量、Kantorovich 度量或土方移动距离（EMD）。在某些条件下，这种度量属于 Integral Probability Metrics (IPM) 的家族（参见第 2 节）。在许多问题（例如生成建模）中，由于其拓扑学、统计学和几何学性质，Wasserstein 距离比其他分布差异的概念（如 Kullback-Leibler (KL) 散度）更优选。其次，OT 提供了一个 *toolkit* 或框架，使机器学习从业者能够操作概率分布。因此，OT 是一个原理性的工具，用于理解概率分布的空间。

这项调查提供了OTML最近演变的最新观点。尽管之前存在[6]、[7]、[8]、[9]、[10]、[11]等调查，但该领域的快速发展证明了对OTML进行更深入审视的必要性。本文组织如下。第2节介绍了OT理论的概述。第3节回顾了 *computational optimal transport*的最新进展。后续章节探讨了OT在4个机器学习问题中的应用：监督学习（第4节）、无监督学习（第5节）、迁移学习（第6节）和强化学习（第7节）。第8节以一般性评论和未来研究方向总结本文。

• *The authors are with the Université Paris-Saclay, CEA, LIST, F-91120, Palaiseau, France. E-mail: eduardo.fernandesmontesuma@cea.fr*

Manuscript received ; revised

2 背景

在下面的内容中，我们简要回顾一下OT在$\mathbb{R}^d$上的应用。对于OT理论的更详细概述，我们建议读者参考[12]。概率分布的空间用$\mathbb{P}(\mathbb{R}^d)$表示。对于一个映射$T: \mathbb{R}^d \rightarrow \mathbb{R}^d$，其相关的*pushforward operator*、$T_\sharp$为，

$$(T_\sharp P)(A) = P(T^{-1}(A)), \text{ for } A \subset \mathbb{R}^d. \tag{1}$$

我们用 Lip_1 表示 1-Lipschitz 函数的集合，并用 $\text{CVX}(P)$ 表示与 $P \in \mathbb{P}(\mathbb{R}^d)$ 有有限矩的凸函数集合。对于 $f: \mathbb{R}^d \rightarrow \mathbb{R}$，*convex conjugate* 为，

$$f^*(\mathbf{x}_2) = \sup_{\mathbf{x}_1 \in \mathbb{R}^d} \langle \mathbf{x}_1, \mathbf{x}_2 \rangle - f(\mathbf{x}_1). \tag{2}$$

令 $P, Q \in \mathbb{P}(\mathbb{R}^d)$。蒙格公式 [13] 寻找最优运输映射 T，使得，

$$\inf_{T_\sharp P = Q} \mathcal{L}_M(T) := \mathop{\mathbb{E}}_{\mathbf{x} \sim P}[c(\mathbf{x}, T(\mathbf{x}))], \tag{3}$$

其中 $c: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_+$ 称为 *ground-cost*。而 $\mathcal{L}_M$ 定义了 *effort of transportation*，$T_\sharp P = Q$ 规定了 *mass conservation*，即 $(T_\sharp P)(A) = Q(A)$ 对于 $A \subset \mathbb{R}^d$。Monge 形式化 notoriously 难以分析，部分原因是由于涉及的约束条件 $T_\sharp$。一个更简单的形式化 [2] 依赖于一个 OT 计划 $\gamma: \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, 1]$，使得，

$$\inf_{\gamma \in \Gamma(P,Q)} \mathcal{L}_K(\gamma) := \mathop{\mathbb{E}}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma}[c(\mathbf{x}_1, \mathbf{x}_2)], \tag{4}$$

其中 $\Gamma(P, Q)$ 是 *mass preserving plans* 的集合，即对于 $A, B \subset \mathbb{R}^d$，$\gamma(\mathbb{R}^d, B) = Q(B)$, 和 , $\gamma(A, \mathbb{R}^d) = P(A)$。这种表述被称为蒙格-卡托瓦奇（MK）。

MK形式化更容易分析，因为$\Gamma(P, Q)$和$\mathcal{L}_K(\gamma)$相对于γ是线性的，这使其成为线性规划。因此，MK形式化在 *Kantorovich potentials* φ方面承认一个对偶问题[12, 第1.2节]：$\mathbb{R}^d \rightarrow \mathbb{R}$和$\psi: \mathbb{R}^d \rightarrow \mathbb{R}$，

$$\sup_{(\varphi,\psi) \in \Phi_c} \mathcal{L}_K^*(\varphi, \psi) := \mathop{\mathbb{E}}_{\mathbf{x} \sim P}[\varphi(\mathbf{x})] + \mathop{\mathbb{E}}_{\mathbf{x} \sim Q}[\psi(\mathbf{x})], \tag{5}$$

其中 $\Phi_c = \{(\varphi,\psi): \varphi(\mathbf{x}_1) + \psi(\mathbf{x}_2) \le c(\mathbf{x}_1,\mathbf{x}_2)\}$。对于 $c(\mathbf{x}_1,\mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|_2^2$，著名的 Brenier 定理 [14] 建立了 eq. 3 和 eq. 4 之间的联系：$T = \nabla\varphi$。

此外，当地面成本为 $c(\mathbf{x}_1,\mathbf{x}_2) = d(\mathbf{x}_1,\mathbf{x}_2)^p$，对于 d 上的度量 $\mathcal{X}$ 和 $p \in [1,\infty)$ 时，OT 假设特殊形式。对于 $p=1$，有 Kantorovich-Rubinstein (KR) 公式 [15, 定理 1.14],

$$\sup_{\varphi\in\mathrm{Lip}_1} \mathcal{L}^*_{KR}(\varphi) := \mathop{\mathbb{E}}_{\mathbf{x}\sim P}[\varphi(\mathbf{x})] - \mathop{\mathbb{E}}_{\mathbf{x}\sim Q}[\varphi(\mathbf{x})], \tag{6}$$

在平行地，对于 $c(\mathbf{x}_1,\mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|_2^p$，$p>1$，可以考虑一个反映质量从 P移动到 Q 的时变版本的OT。这被称为动态OT [12, 第6章]，并用时变分布 $\rho(t,\mathbf{x})$ 表述，其中 $\rho(0,\cdot) = P$且 $\rho(1,\cdot) = Q$，以及一个向量场 $\mathbf{v}$ 定义质量如何移动。用这些术语，eq. 4 变为,

$$\mathcal{L}_B(\rho,\mathbf{v}) := \int_0^1\int_{\mathbb{R}^d} \|\mathbf{v}(t,\mathbf{x})\|_2^p \rho_t(\mathbf{x}) d\mathbf{x}dt, \tag{7}$$

对于 $\rho_t = \rho(t,\cdot)$，在质量守恒约束下,

$$\frac{\partial\rho_t}{\partial t} + \nabla\cdot(\rho_t\mathbf{v}) = 0. \tag{8}$$

我们在图1(a)、(b)和(c)中展示了蒙格、卡纳托维奇和动态OT形式之间的概念比较。最重要的是，由于其多种形式，OT理论是一个相当灵活的工具箱，用于分析概率模型，因此它的流行程度很高。

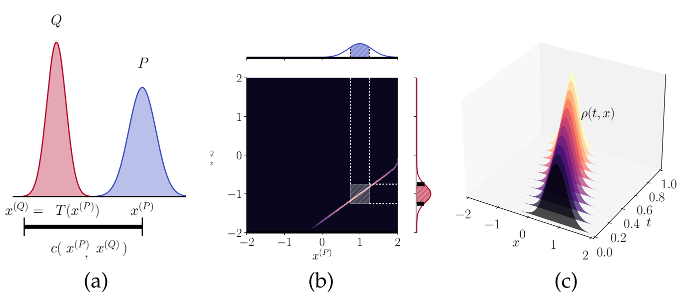

Fig. 1: (a) Monge 形式化，(b) Kantorovich 形式化和 (c) Benamou-Brenier 形式化示意图。其中 (a) 专注于运输映射 T，(b) 依赖于运输计划 γ，而 (c) 围绕插值 $\rho(t,x)$。

OT理论的一个中心方面是，可以根据其解来定义分布之间的损失，即{\(v^*\)}。

$$\mathcal{T}_c(P,Q) = \inf_{\gamma\in\Gamma(P,Q)} \mathop{\mathbb{E}}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}[c(\mathbf{x}_1,\mathbf{x}_2)],$$

这是一种距离，表示 P和 Q 之间的距离。结果发现，当 c 来自 $\mathcal{X}$ 上的度量时，$\mathcal{T}_c$ 本身也成为了一个度量 [12, 第5章]。对于 $p\in[1,\infty]$，有 *Wasserstein distances* 的概念,

$$W_p(P,Q) = (\mathcal{T}_{d^p}(P,Q))^{1/p}, \tag{9}$$

这些在机器学习中广泛使用。这种定义有一些有趣的后果。例如，OT 提供了一种通过 Wasserstein 地线 [16] 和重心 [17] 原则性地插值和平均分布的方法。

关于测地线，如[3，第7章]所示，测地线从P到Q被定义为一个分布 $P_t = \pi_{t,\sharp}\gamma$，其中 $\pi_t(\mathbf{x}_1,\mathbf{x}_2) = (1-t)\mathbf{x}_1 + t\mathbf{x}_2$。类似地，令 $\mathcal{P} = \{P_i\}_{i=1}^N$，为 $\mathcal{P}$ 的 Wasserstein 广义中心 [17]，权重为 $\alpha\in\Delta_N = \{\mathbf{a}\in\mathbb{R}_+^N : \sum_{i=1}^N a_i = 1\}$,

$$\mathcal{B}(\alpha;\mathcal{P}) = \mathop{\arg\inf}_{Q} \sum_{i=1}^N \alpha_i W_p(P_i,Q)^p. \tag{10}$$

Probability metrics 都是量化两个概率分布差异的泛函。这些可以是适当的度量（例如，Wasserstein 距离）或散度（例如，KL 散度）。在概率论中，有两个 prominant 的度量家族，积分概率度量（IPMs）[18] 和 $f-$ 散度 [19]。对于一组函数 $\mathcal{F}$，一个 IPM [20] 给出为,

$$d_{\mathcal{F}}(P,Q) = \sup_{f\in\mathcal{F}} \left| \mathop{\mathbb{E}}_{\mathbf{x}\sim P}[f(\mathbf{x})] - \mathop{\mathbb{E}}_{\mathbf{x}\sim Q}[f(\mathbf{x})] \right|.$$

因此，IPMs 根据分布之间的差异 $P-Q$ 来衡量距离。由于等式 6，W_1 是一个具有 $\mathcal{F}$ =Lip_1 的 IPM，且满足 $c(\mathbf{x}_1,\mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|_2$。另一个重要的度量是最大均值偏差 (MMD) [21]，对于 $\mathcal{F} = \{f\in\mathcal{H}_k$，定义为 $\|f\|_{\mathcal{H}_k}\le 1\}$，其中 $\mathcal{H}_k$ 是一个再生核希尔伯特空间 (RKHS)，带有核 $k:\mathbb{R}^d\times\mathbb{R}^d\to\mathbb{R}$。这些度量在生成建模和域适应中扮演重要角色。相反，$f-$ 散度基于 P和 Q 之间的比率来衡量差异。对于一个凸的下半连续函数 $f:\mathbb{R}_+\to\mathbb{R}$，且 $f(1)=0$，

$$D_f(P||Q) = \mathop{\mathbb{E}}_{\mathbf{x}\sim Q}\left[f\left(\frac{P(\mathbf{x})}{Q(\mathbf{x})}\right)\right], \tag{11}$$

其中，用了一种符号上的 abuse，$P(\mathbf{x})$ (resp. Q) 表示 P的密度。$f-$ 散度的一个例子是 KL 散度，其中 $f(u) = u\log u$。

如[20]所述，有两种属性使IPMs优于f-散度。首先，当P和Q具有不交支持时，$d_{\mathcal{F}}$仍然被定义。例如，在训练的开始阶段，生成对抗网络（GANs）生成质量较差的样本，因此P_{model}和P_{data}具有不交支持。从这个意义上说，IPMs 提供了一个有意义的度量，而 $D_f(P_{model}||P_{data}) = +\infty$则不然，无论$P_{model}$多么糟糕。其次，IPMs 考虑了样本所在的空间的几何结构。例如，考虑高斯分布流形$\mathcal{M} = \{\mathcal{N}(\mu,\sigma^2): \mu\in\mathbb{R},\sigma\in\mathbb{R}_+\}$。如[10, 附注8.2]所述，当限制在$\mathcal{M}$时，具有欧几里得基础成本的Wasserstein距离为,

$$W_2(P,Q) = \sqrt{(\mu_P-\mu_Q)^2 + (\sigma_P-\sigma_Q)^2},$$

而KL散度与双曲几何相关。这如图2所示。总体而言，分布间差异性的选择极大地影响了学习算法的成功（例如，GANs）。确实，每种度量/散度的选择会在概率分布的空间中诱导不同的几何结构，从而改变机器学习中的基础优化问题。

3 计算最优运输

计算OT是ML领域内的一个活跃研究方向。我们建议读者参见[9]、[10]了解其基础，参见[22]、[23]了解广泛使用的软件。在本文综述中，我们专注于

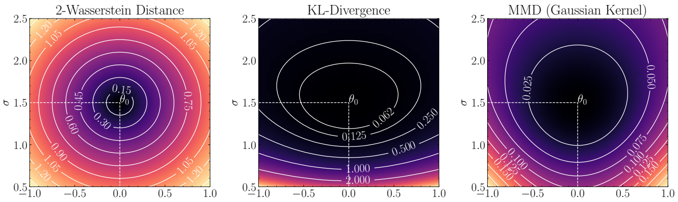

Fig. 2: Comparison on how different metrics and divergences calculate discrepancies on the manifold of Gaussian distributions. The geometry induced by the Wasserstein distance is simpler, and more intuitive than those given by other measures of discrepancy, which affects optimization procedures in machine learning.

two discretization strategies: (i) discretizing the ambient space; (ii) approximating the distributions from samples. In both cases, let $\mathbf{x}_i^{(P)} \sim P$ with probability $p_i > 0$. The *empirical approximation* $\hat{P}$ of P is,

$$\hat{P}(\mathbf{x}) = \sum_{i=1}^{n} p_i \delta(\mathbf{x} - \mathbf{x}_i^{(P)}). \tag{12}$$

Naturally, $\sum_{i=1}^{n} p_i = 1$ or $\mathbf{p} \in \Delta_n$ in short. As follows, discretizing the ambient space is equivalent to binning it, thus assuming a fixed grid of points $\mathbf{x}_i^{(P)}$. The sample weights p_i correspond to the number of points that are assigned to the i-th bin. In this sense, the parameters of $\hat{P}$ are the weights p_i. Conversely, one can sample $\mathbf{x}_i^{(P)} \overset{\text{i.i.d.}}{\sim} P$ (resp. Q). In this case, $p_i = 1/n$, and the parameters of $\hat{P}$ are the locations $\mathbf{x}_i^{(P)}$. We now discuss discrete OT.

Let $\{\mathbf{x}_i^{(P)}\}_{i=1}^{n}$ (resp. $\{\mathbf{x}_j^{(Q)}\}_{j=1}^{m}$) sampled from P (resp. Q) with probability p_i (resp. q_j). The Monge problem seeks a mapping T, that is the solution of,

$$T^{\star} = \underset{T_{\sharp}\hat{P}=\hat{Q}}{\arg\min}\, \mathcal{L}_M(T) = \sum_{i=1}^{n} c(\mathbf{x}_i^{(P)}, T(\mathbf{x}_i^{(P)})), \tag{13}$$

where the constraint implies $\sum_{i \in \mathcal{I}} p_i = q_j$, for $\mathcal{I} = \{i : \mathbf{x}_j^{(Q)} = T(\mathbf{x}_i^{(P)})\}$. This formulation is non-linear w.r.t. T. In addition, for $m > n$, it does not have a solution. Conversely, the MK formulation seeks an OT *plan* $\gamma \in \mathbb{R}^{n \times m}$, where γ_{ij} denotes the amount of mass transported from sample i to sample j. In this case γ must minimize,

$$\hat{\gamma} = \underset{\gamma \in \Gamma(\mathbf{p},\mathbf{q})}{\arg\min}\, \mathcal{L}_K(\gamma) = \sum_{i=1}^{n}\sum_{j=1}^{m} \gamma_{ij} c(\mathbf{x}_i^{(P)}, \mathbf{x}_j^{(Q)}), \tag{14}$$

where $\Gamma(\mathbf{p},\mathbf{q}) = \{\gamma \in \mathbb{R}^{n \times m} : \sum_i \gamma_{ij} = q_j \text{ and } \sum_j \gamma_{ij} = p_i\}$. This is a linear program, which can be solved through the Simplex algorithm [24], with time complexity $\mathcal{O}(n^3 \log n)$. Alternatively, one can use the approximation introduced by [25], by solving a regularized problem,

$$\hat{\gamma}_\epsilon = \underset{\gamma \in \Gamma(\mathbf{p},\mathbf{q})}{\arg\min} \sum_{i=1}^{n}\sum_{j=1}^{m} \gamma_{ij} c(\mathbf{x}_i^{(P)}, \mathbf{x}_j^{(Q)}) + \epsilon H(\gamma), \tag{15}$$

which provides a faster way to estimate γ. An additional advantage of the Sinkhorn algorithm is that

$$\hat{\gamma}_\epsilon = \text{diag}(\mathbf{f}) e^{-\mathbf{C}/\epsilon} \text{diag}(\mathbf{g}), \tag{16}$$

where, as in eq. 5, $(\mathbf{f}, \mathbf{g})$ are the Kantorovich potentials.

Solving OT with finite samples provides an empirical estimator for $\mathcal{T}_c$ and W_p, i.e., $\mathcal{T}_c(\hat{P}, \hat{Q}) = \mathcal{L}_K(\hat{\gamma})$. Likewise, for $\gamma_\epsilon^\star$ one has $\mathcal{T}_{c,\epsilon}(\hat{P}, \hat{Q}) = \mathcal{L}_K(\gamma_\epsilon^\star)$. This approximation motivates the Sinkhorn divergence [26],

$$S_{p,\epsilon}(\hat{P}, \hat{Q}) = W_{p,\epsilon}(\hat{P}, \hat{Q}) - \frac{W_{p,\epsilon}(\hat{P}, \hat{P}) + W_{p,\epsilon}(\hat{Q}, \hat{Q})}{2}, \tag{17}$$

which has interesting properties, such as interpolating between the MMD of [21] and the Wasserstein distance. Overall, entropic OT has two computational advantages w.r.t exact OT. Indeed, its calculations are GPU-friendly, and for $L \geq 1$ iterations, its complexity is $\mathcal{O}(Ln^2)$. In addition, $S_{c,\epsilon}$ is a smooth approximator of W_p [27], and it enjoys better sample complexity [28] (c.f., section 8.1).

In the following, we discuss recent innovations on computational OT. Section 3.1 present projection-based methods. Section 3.2 discusses OT formulations with prescribed structures. Section 3.3 presents OT through Input Convex Neural Networks (ICNNs). Section 3.4 explores how to compute OT between mini-batches of data.

3.1 Projection-based Optimal Transport

Projection-based OT relies on projecting data $\mathbf{x} \in \mathbb{R}^d$ into sub-spaces $\mathbb{R}^k$, $k < d$. A natural choice is $k = 1$, for which computing OT can be done by sorting [12, chapter 2]. This is called Sliced Wasserstein (SW) distance [29], [30]. Let $\mathbb{S}^{d-1} = \{\mathbf{u} \in \mathbb{R}^d : \|\mathbf{u}\|_2^2 = 1\}$ denote the unit-sphere in $\mathbb{R}^d$, and $\pi_{\mathbf{u}} : \mathbb{R}^d \to \mathbb{R}$ denote $\pi_{\mathbf{u}}(\mathbf{x}) = \langle \mathbf{u}, \mathbf{x} \rangle$. The Sliced-Wasserstein distance is,

$$\text{SW}_p(P, Q)^p = \int_{\mathbb{S}^{d-1}} W_p^p(\pi_{\mathbf{u},\sharp}\hat{P}, \pi_{\mathbf{u},\sharp}\hat{Q}) d\mathbf{u}. \tag{18}$$

We highlight a few advantages. First, $W_p(\pi_{\mathbf{u},\sharp}P, \pi_{\mathbf{u},\sharp}Q)$ can be computed in $\mathcal{O}(n \log n)$ [10]. Second, the integration in equation 18 can be computed using Monte Carlo estimation. For samples $\{\mathbf{u}_\ell\}_{\ell=1}^{L}$, $\mathbf{u}_\ell \in \mathbb{S}^{d-1}$ uniformly,

$$\text{SW}_p(\hat{P}, \hat{Q})^p = \frac{1}{L}\sum_{\ell=1}^{L} W_p^p(\pi_{\mathbf{u}_\ell,\sharp}\hat{P}, \pi_{\mathbf{u}_\ell,\sharp}\hat{Q}), \tag{19}$$

which implies that SW(P, Q) has $\mathcal{O}(Lnd + Ln \log n)$ time complexity. As shown in [31], $SW(P, Q)$ is indeed a metric. In addition, [32] and [33] proposed variants of SW, namely, the max-SW distance and the generalized SW distance respectively. Contrary to the averaging procedure in eq. 18, the max-SW of [32] takes the direction with maximum distance between P and Q,

$$\text{max-SW}_p^p(P, Q) = \max_{\mathbf{u} \in \mathbb{S}^{d-1}} W_p^p(\pi_{\mathbf{u},\sharp}P, \pi_{\mathbf{u},\sharp}Q). \tag{20}$$

This metric has the same advantage in sample complexity as the SW distance, while being easier to compute. We illustrate these concepts in Figure 3 for $P, Q \in \mathbb{P}(\mathbb{R}^2)$.

With respect to figure 3, note that projection directions are not equally important. This is illustrated in 3 (d), in which some directions have higher distance than others. This phenomenon was analyzed by [34], who proposed the so-called Distributional SW (DSW) distance,

$$\text{DSW}_p(P, Q; C) = \sup_{\sigma \in \mathbb{M}_C} \left(\mathop{\mathbb{E}}_{\mathbf{u} \sim \sigma} [W_p^p(\pi_{\mathbf{u},\sharp}P, \pi_{\mathbf{u},\sharp}Q)] \right)^{1/p},$$

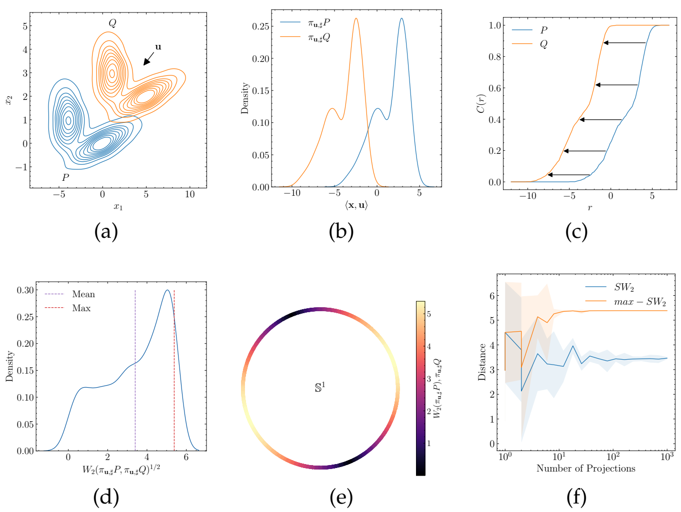

Fig. 3: An illustration of the sliced and max-sliced Wasserstein distances over 2-D distributions (a). In (b), we show the densities of P and Q after a projection by $\mathbf{u}$. In (c), we illustrate the computation of the 1-D Wasserstein distance for $p = 1$, as the horizontal difference between the cumulative distributions of P and Q. In (d), we show the distribution of the Wasserstein distance over $\mathbf{u} \sim \mathbb{S}^1$, alongside the mean (purple) and max (red) values. In (e), we show the Wasserstein distances over $\mathbf{u} \in \mathbb{S}^1$. Finally, (f) shows the estimation of the SW_2 and max-SW_2 as a function of the number of projections L. Shaded regions show a 95% confidence interval around the average value.

for a family $\mathbb{M}_C = \{\sigma \in \mathbb{P}(\mathbb{S}^{d-1}) : \mathbb{E}_{\mathbf{u},\mathbf{u}'\sim\sigma}[|\mathbf{u}^T\mathbf{u}'|] \leq C\}$. The intuition behind this metric is that σ weights the directions sampled from $\mathbb{S}^{d-1}$.

In addition, one can project samples on a sub-space $1 < k < d$. For instance, [35] proposed the Subspace Robust Wasserstein (SRW) distances,

$$\text{SRW}_k(P,Q)^2 = \inf_{\gamma\in\Gamma(P,Q)} \sup_{E\in\mathcal{G}_k} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}[\|\pi_E(\mathbf{x}_1 - \mathbf{x}_2)\|_2^2],$$

where $\mathcal{G}_k = \{E \subset \mathbb{R}^d : dim(E) = k\}$ is the Grassmannian manifold of $k-$dimensional subspaces of $\mathbb{R}^d$, and π_E denote the orthogonal projector onto E. This can be equivalently formulated through a projection matrix $\mathbf{U} \in \mathbb{R}^{k\times d}$, that is,

$$\text{SRW}_k(P,Q)^2 = \inf_{\Gamma(P,Q)} \max_{\substack{\mathbf{U}\in\mathbb{R}^{k\times d}\\ \mathbf{U}\mathbf{U}^T=\mathbf{I}_k}} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}[\|\mathbf{U}\mathbf{x}_1 - \mathbf{U}\mathbf{x}_2\|_2^2]$$

In practice, SRW_k is based on a projected super gradient method [35, Algorithm 1] that updates $\Omega = \mathbf{U}\mathbf{U}^T$, for a fixed γ, until convergence. These updates are computationally complex, as they rely on eigendecomposition. Further developments using Riemannian [36] and Block Coordinate Descent (BCD) [37] circumvent this issue by optimizing over $\text{St}(d,k) = \{\mathbf{U} \in \mathbb{R}^{k\times d} : \mathbf{U}\mathbf{U}^T = \mathbf{I}_k\}$.

3.2 Structured Optimal Transport

In some cases, it is desirable for the OT plan to have additional structure, e.g., in color transfer [38] and domain adaptation [39]. Based on this problem, [40] introduced a principled way to compute structured OT through sub-modular costs.

As defined in [40], a set function $F : 2^V \rightarrow \mathbb{R}$ is sub-modular if $\forall S \subset T \subset V$ and $\forall v \in V\backslash T$,

$$F(S \cup \{v\}) - F(S) \geq F(T \cup \{v\}) - F(T).$$

These types of functions arise in combinatorial optimization, and further OT since OT mappings and plans can be seen as a matching between 2 sets, namely, samples from P and Q. In addition, F defines a *base polytope*,

$$\mathcal{B}_F = \{G \in \mathbb{R}^{|V|} : G(V) = F(V); G(S) \leq F(S)\; \forall S \subset V\}.$$

Based on these concepts, note that OT can be formulated in terms of set functions. Indeed, suppose $\mathbf{X}^{(P)} \in \mathbb{R}^{n\times d}$ and $\mathbf{X}^{(Q)} \in \mathbb{R}^{m\times d}$. In this case, $\gamma^\star$ or $T^\star$ can be interpreted as a graph with edge set $E = \{(u_\ell, v_\ell)\}_{\ell=1}^k$, where u_ℓ represents a sample in P and v_ℓ, the corresponding sample (through γ) in Q. In this case, the cost of transportation is represented by $F(S) = \sum_{(u,v)\in S} c_{uv}$. Hence, adding a new (u, v) to S is the same regardless of the elements of S.

The insight of [40] is using the sub-modular property for acquiring structured OT plans. Through Lovász extension [41], this leads to,

$$(\gamma^\star, \kappa^\star) = \operatorname*{arg\,min}_{\gamma\in\Gamma(\mathbf{p},\mathbf{q})} \operatorname*{arg\,max}_{\kappa\in\mathcal{B}_F} \langle\gamma, \kappa\rangle_F.$$

A similar direction was explored by [42], who proposed to impose a low rank structure on OT plans. This was done with the purpose of tackling the curse of dimensionality in OT. They introduce the transport rank of $\gamma \in \Gamma(P,Q)$ defined as the smallest integer K for which,

$$\gamma = \sum_{k=1}^K \lambda_k (P_k \otimes Q_k),$$

where $P_k, Q_k, k = 1, \cdots, K$ are distributions over $\mathbb{R}^d$, and $P_k \otimes Q_k$ denotes the (independent) joint distribution with marginals P_k and Q_k, i.e. $(P_k \otimes Q_k)(\mathbf{x},\mathbf{y}) = P_k(\mathbf{x})Q_k(\mathbf{y})$. For empirical $\hat{P}$ and $\hat{Q}$, K coincides with the non-negative rank [43] of $\gamma \in \mathbb{R}^{n\times m}$. As follows, [42] denotes the set of γ with transport rank at most K as $\Gamma_K(P,Q)$ ($\Gamma_K(\mathbf{p},\mathbf{q})$ for empirical $\hat{P}$ and $\hat{Q}$). The robust Wasserstein distance is thus,

$$\text{FW}_{K,2}(P,Q)^2 = \inf_{\gamma\in\Gamma_K(P,Q)} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x})\sim\gamma}[\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2].$$

In practice, this optimization problem is difficult due to the constraint $\gamma \in \Gamma_K$. As follows, [42] propose estimating it using the Wasserstein barycenter $B = \mathcal{B}([1/2, 1/2]; \{P,Q\})$ supported on K points, that is $\{\mathbf{x}_k^{(B)}\}_{k=1}^K$, also called *hubs*. As follows, the authors show how to construct a transport plan in $\Gamma_k(P,Q)$, by exploiting the transport plans $\gamma_1 \in \Gamma(P,B)$ and $\gamma_2 \in \Gamma(B,Q)$. This establishes a link between the robustness of the Wasserstein distance, Wasserstein barycenters and clustering. Let $\lambda_k = \sum_{i=1}^n \gamma_{ki}^{(BP)}$ and $\mu_k^{(P)} = 1/\lambda_k \sum_{i=1}^n \gamma_{ki}^{(BP)}\mathbf{x}_i^{(P)}$, the authors in [42] propose the following proxy for the Wasserstein distance,

$$\text{FW}_{k,2}^2(\hat{P},\hat{Q})^2 = \sum_{k=1}^K \lambda_k \|\mu_k^{(P)} - \mu_k^{(Q)}\|_2.$$

3.3 Neural Network-based Solvers

In ML, different works estimate OT through Neural Networks (NNs) [44], [45], [46], [47]. For instance, as we cover in section 5.1, [44] proposes to estimate the Kantorovich-Rubinstein distance in eq. 6 by parametrizing φ through a NN. In the following we cover how different works approximate OT plans and maps through NNs.

Neural OT Plans. [45] was the first to propose to approximate OT plans through NNs. Their approach relies on solving the entropic regularized dual Kantorovich problem in eq. 5, by parametrizing φ and ψ through NNs (u_ξ, v_η). The optimization procedure consists on maximizing,

$$\sup_{\xi,\eta} \underset{\mathbf{x}_1\sim P,\mathbf{x}_2\sim Q}{\mathbb{E}}[u_\xi(\mathbf{x}_1)+v_\eta(\mathbf{x}_2)-\epsilon\gamma_\epsilon(\mathbf{x}_1,\mathbf{x}_2)],$$

where, in analogy with eq. 16,

$$\gamma_\epsilon(\mathbf{x}_1,\mathbf{x}_2)=e^{-(u_\xi(\mathbf{x}_1)+v_\eta(\mathbf{x}_2)-c(\mathbf{x}_1,\mathbf{x}_2))/\epsilon}.$$

As a result, for optimal (ξ,η), one can estimate the OT plan for pairs $\mathbf{x}_1\sim P$ and $\mathbf{x}_2\sim Q$.

Neural OT Maps. As we discussed in the previous topic, [45] proposed a way to approximate entropic regularized Kantorovich potentials. Based on this optimization procedure, the authors propose approximating the so-called barycentric mapping through a NN $f_\theta(\mathbf{x})$,

$$\min_\theta \underset{\mathbf{x}_1\sim P,\mathbf{x}_2\sim Q}{\mathbb{E}}[d(f_\theta(\mathbf{x}_1),\mathbf{x}_2)\gamma_\epsilon(\mathbf{x}_1,\mathbf{x}_2)],$$

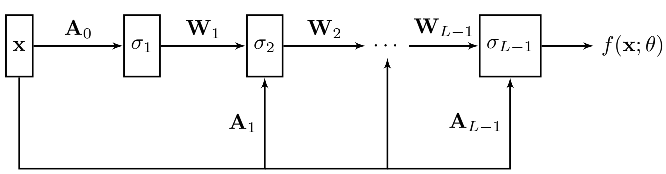

Fig. 4: ICNN architecture proposed by [48], which implements a convex function $f(\mathbf{x};\theta)$ with respect inputs $\mathbf{x}$.

A further development, when $c(\mathbf{x}_1,\mathbf{x}_2)=\|\mathbf{x}_1-\mathbf{x}_2\|_2^2$, was proposed by [46], who relies on convex analysis [15] and ICNNs [48]. Formally, an ICNN implements a function $f:\mathbb{R}^d\rightarrow\mathbb{R}$ such that $\mathbf{x}\mapsto f_\theta(\mathbf{x})$ is convex. This is achieved through a special structure, shown in figure 4. An $L-$layer ICNN is defined through the operations,

$$\mathbf{z}_{\ell+1}=\sigma_\ell(\mathbf{W}_\ell\mathbf{z}_\ell+\mathbf{A}_\ell\mathbf{x}+\mathbf{b}_\ell) \text{ and } f_\theta(\mathbf{x})=\mathbf{z}_L,$$

where all entries of $\mathbf{W}_\ell$ are non-negative, σ_0 is convex and $\sigma_\ell, \ell=1,\cdots,L-1$ is convex and non-decreasing. The ICNN architecture is shown in Figure 4.

The insight from [46] comes from rewriting eq. 5 as a mini-max problem involving two convex functions $f_\theta\in\text{CVX}(P)$ and $g_\eta\in\text{CVX}(Q)$,

$$W_2(P,Q)^2=\sup_\theta\inf_\eta\mathcal{V}(\theta,\eta)+\mathcal{E}, \tag{21}$$
$$\mathcal{V}(\theta,\eta)=-\underset{\mathbf{x}\sim P}{\mathbb{E}}[f_\theta(\mathbf{x})]-\underset{\mathbf{x}\sim Q}{\mathbb{E}}[\langle\mathbf{x},\nabla g_\eta(\mathbf{x})\rangle-f_\theta(\nabla g_\eta(\mathbf{x}))],$$
$$\mathcal{E}=\mathbb{E}_{\mathbf{x}\sim P}[\|\mathbf{x}\|_2^2/2]+\mathbb{E}_{\mathbf{x}\sim Q}[\|\mathbf{x}\|_2^2/2].$$

As shown by the authors, this mini-max problem can be approximated empirically from samples $\mathbf{x}_i^{(P)}\sim P$ and $\mathbf{x}_i^{(Q)}\sim Q$. Furthermore, due to Brenier's theorem [14], $T=\nabla_{\mathbf{x}}g_\eta$. As a consequence, one is able to find an OT map by taking the gradient $\nabla_{\mathbf{x}}g_\eta(\mathbf{x})$.

3.4 Mini-batch Optimal Transport

A major challenge in OT is its time complexity. This motivated different authors [26], [49], [50] to compute the Wasserstein distance between mini-batches rather than complete datasets. For a dataset with n samples, this strategy leads to a dramatic speed-up, since for K mini-batches of size $m\ll n$, one reduces the time complexity of OT from $\mathcal{O}(n^3\log n)$ to $\mathcal{O}(Km^3\log m)$ [50]. This choice is key when using OT as a loss in learning [51] and inference [52]. Henceforth we describe the mini-batch framework of [53], for using OT as a loss.

Let $\mathcal{L}_{OT}$ denote an OT loss (e.g., W_p or $S_{p,\epsilon}$). Assuming continuous distributions P and Q, the Mini-batch OT (MBOT) loss is given by,

$$\mathcal{L}_{\text{MBOT}}(P,Q)=\underset{(\mathbf{X}^{(P)},\mathbf{X}^{(Q)})\sim P^{\otimes m}\otimes Q^{\otimes m}}{\mathbb{E}}[\mathcal{L}_{OT}(\mathbf{X}^{(P)},\mathbf{X}^{(Q)})],$$

where $\mathbf{X}^{(P)}\sim P^{\otimes m}$ indicates $\mathbf{x}_i^{(P)}\sim P, i=1,\cdots,m$. This loss inherits some properties from OT, i.e., it is positive and symmetric, but $\mathcal{L}_{\text{MBOT}}(P,P)>0$. In practice, let $\{\mathbf{x}_i^{(P)}\}_{i=1}^{n_P}$ and $\{\mathbf{x}_j^{(Q)}\}_{i=1}^{n_Q}$ be iid samples from P and Q respectively. Let $\mathcal{I}_m\subset\{1,\cdots,n_P\}^m$ denote a set of m indices. We denote by $\hat{P}_{\mathcal{I}_m}$ to the empirical approximation of P with a single mini-batch: $\mathbf{X}^{(P)}=\{\mathbf{x}_i^{(P)}:i\in\mathcal{I}_m\}$. Therefore,

$$\mathcal{L}_{\text{MBOT}}^{(k,m)}(\hat{P},\hat{Q})=\frac{1}{k}\sum_{(\mathcal{I}_b,\mathcal{I}'_b)\in\mathbb{I}_k}\mathcal{L}_{OT}(\hat{P}_{\mathcal{I}_b},\hat{Q}_{\mathcal{I}'_b}), \tag{22}$$

where $\mathbb{I}_k$ is a random set of k mini-batches of size m from P and Q. This constitutes an estimator for $\mathcal{L}_{MBOT}(P,Q)$, which converges as n and $k\rightarrow\infty$. We highlight 3 advantages that favor MBOT for ML: (i) it is faster to compute and computationally scalable; (ii) the deviation bound between $\mathcal{L}_{\text{MBOT}}(P,Q)$ and $\mathcal{L}_{\text{MBOT}}^{(k,m)}$ does not depend on the dimensionality of the space; (iii) it has unbiased gradients, i.e., the expected gradient of the sample loss equals the gradient of the true loss. Nonetheless, mini-batch OT brings new challenges. As [54] studies, the use of mini-batches introduces artifacts in OT plans, as they become less sparse. This issue is shown in Figure 5, which shows the OT plan in mini-batch OT. We provide further discussion in the next section.

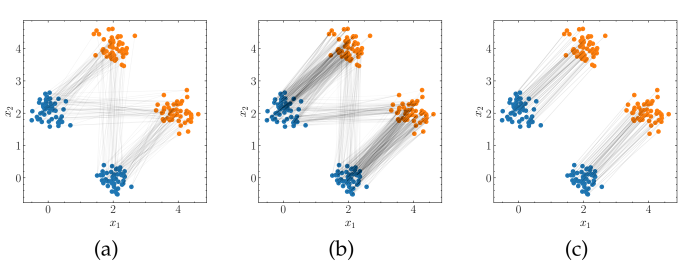

Fig. 5: Mini-batch OT between distributions P (in blue) and Q (in orange). As follows, an OT plan is calculated with mini-batches of 2 (a), 10 (b) and 100 (c) samples. (c) corresponds to the original OT problem. Overall, in mini-batch OT the plans become less sparse, due to OT being forced to transport all mass between mini-batches.

3.5 Extensions to Optimal Transport

In this section, we cover 3 extensions to the OT problem, namely: (i) unbalanced OT (ii) partial OT and (iii) OT between incomparable spaces.

Unbalanced OT is an extension to the original Kantorovich problem, which relaxes the mass conservation constraint [53]. The idea, as discussed in [55], is to replace the hard constraint $\gamma \in \Gamma(\mathbf{p}, \mathbf{q})$ by soft constraints in terms of a f-divergence (cf., eq. 11),

$$\hat{\gamma}_{\epsilon,\tau} = \arg\min_{\gamma} \sum_{i=1}^{n}\sum_{j=1}^{m}\gamma_{ij}c(\mathbf{x}_i^{(P)}, \mathbf{x}_j^{(Q)}) + \epsilon H(\gamma) + \tau(D_f(\gamma_1|\mathbf{p}) + D_f(\gamma_2|\mathbf{q})). \tag{23}$$

In analogy with the Sinkhorn divergence, unbalanced OT defines a divergence $S_{\epsilon,\tau}$ as well. This extension has a few advantages. First, it can be easily implemented on top of the Sinkhorn algorithm [56]. Second, it is well defined for positive vectors $\mathbf{p} \in \mathbb{R}^n_+$, $\mathbf{q} \in \mathbb{R}^m_+$. Third, it is robust to outliers [54], which favors its application to mini-batch OT.

Partial OT defines an OT problem in which the transportation plan do not transport a fraction, $0 \leq s \leq 1$, of the total mass. This defines a new set,

$$\Gamma_s(\mathbf{p}, \mathbf{q}) = \{\gamma : \sum_i \gamma_{ij} \leq q_j, \sum_j \gamma_{ij} \leq p_i, \sum_{i,j}\gamma_{ij} = s\},$$

which substitutes Γ in eq. 14. As [57] proposes, partial OT can be solved by adding dummy sink points to which the mass that is not transported, $1 - s$, will be sent to. Similarly to unbalanced OT, the partial extension is used to enhance mini-batch OT [58].

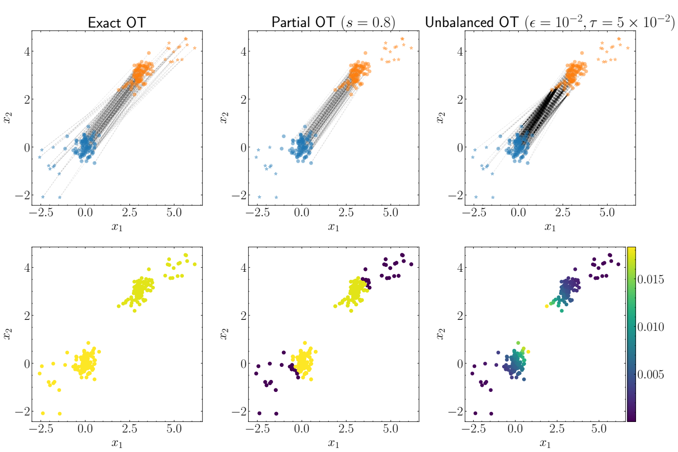

Fig. 6: Comparison between exact, partial and unbalanced OT. On top, we visualize the OT plans as lines joining points transporting mass. On bottom, we color samples by how much mass they send (source distribution) and receive (target distribution), which shows that most outliers in the distributions do not participate in partial nor unbalanced OT. This phenomenon highlights the advantage of these extensions for handling datasets with outliers.

Gromov-Wasserstein OT constitutes a series of theoretical extensions to the OT problem, when P and Q are probability distributions over *different spaces*. As a consequence, one cannot compute $C_{ij} = c(\mathbf{x}_i^{(P)}, \mathbf{x}_j^{(Q)})$. To that end, one can introduce the Gromov-Wasserstein (GW) distance [59],

$$\text{GW}(P, Q) = \min_{\gamma \in \Gamma(\mathbf{p},\mathbf{q})} \sum_{i,j,k,l} \mathcal{L}(S_{i,k}^{(P)}, S_{j,l}^{(Q)})\gamma_{i,k}\gamma_{j,l}, \tag{24}$$

where $S_{i,j}^{(P)}$ quantify the similarity between objects i and j in P (resp. Q). We can illustrate this idea with graphs. Formally, a graph $G = (V, E)$ is a pair of a set of vertices E and a set of edges E. A histogram over a graph is a pair $P = (V^{(P)}, E^{(P)}, \mathbf{p})$, in which p_i is the weight of vertex i. An example of Gromov-Wasserstein distance over graph histograms is considering $S_{i,j}^{(P)}$ as the shortest-path distance between vertices i and j.

Additionally, one can define structured objects over graphs, by assigning features to vertices [60], i.e., $\hat{P} = (V^{(P)}, E^{(P)}, \mathbf{X}^{(P)}, \mathbf{p})$, where $\mathbf{X}^{(P)} \in \mathbb{R}^{|V|\times d}$. In this context, [60] proposed the Fused Gromov-Wasserstein (FGW) distance, which interpolates between the standard Wasserstein distance between $(\mathbf{p}, \mathbf{X}^{(P)})$ and $(\mathbf{q}, \mathbf{X}^{(Q)})$, and the GW distance between $(V^{(P)}, E^{(P)}, \mathbf{p})$ and $(V^{(Q)}, E^{(Q)}, \mathbf{q})$, i.e.,

$$\text{FGW}_\alpha(\hat{P}, \hat{Q}) = \min_{\gamma \in \Gamma(\mathbf{p},\mathbf{q})} \sum_{i,j,k,l} L_{i,j,k,l}\gamma_{i,k}\gamma_{j,l}, \quad \text{for } L_{i,j,k,l} = (1-\alpha)c(\mathbf{x}_i^{(P)}, \mathbf{x}_j^{(Q)}) + \alpha\mathcal{L}(S_{i,k}^{(P)}, S_{j,l}^{(Q)}). \tag{25}$$

The ideas behind eqs. 24 and 25 are illustrated in 7, in which two graphs are compared with the GW and FGW distances. In contrast with standard OT, the Gromov extension of OT is notoriously harder to solve, as problems in eqs. 24 and 25 are non-convex. Especially, eq. 24 is equivalent to a quadratic assignment problem [61], which has NP-hard complexity for arbitrary inputs [10, Section 10.6]. Like standard OT, one can add entropic regularization to eq. 24 [62], [63], alleviating its computational complexity.

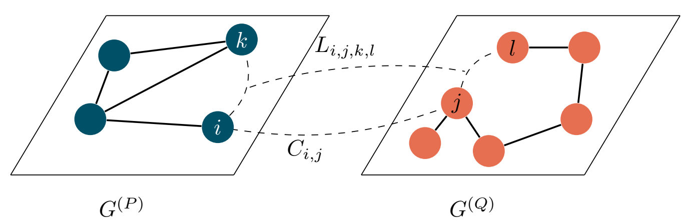

Fig. 7: Comparison between two graphs $G^{(P)}$ and $G^{(Q)}$ which lie in different spaces. The Gromov extension of OT consists on comparing links (i, k) and (k, l) in both graphs. When features are available for vertices, one can add a cost $C_{ij} = c(\mathbf{x}_i^{(P)}, \mathbf{x}_j^{(Q)})$, leading to the FGW distance of [60].

Weak Optimal Transport [64] is a generalization of OT, in which the ground-cost $c : \mathbb{R}^d \times \mathbb{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ measures the effort of transportation between a point $\mathbf{x}_1 \sim P$, and a distribution conditional $\gamma(\cdot|\mathbf{x}_1)$. In analogy with eq. 14,

$$\mathcal{T}_c(P, Q) = \inf_{\gamma \in \Gamma(P,Q)} \mathbb{E}_{\mathbf{x}_1 \sim P}[c(\mathbf{x}_1, \gamma(\cdot|\mathbf{x}_1))]. \tag{26}$$

The original (strong) OT formulation may be retrieved by considering the cost $c(\mathbf{x}_1, Q) = \mathbb{E}_{\mathbf{x}_2 \sim Q}[c(\mathbf{x}_1, \mathbf{x}_2)]$. In a similar way to Kantorovich duality (eq. 5), weak-OT has a dual formulation as well,

$$\mathcal{T}_c(P, Q) = \sup_f \mathbb{E}_{\mathbf{x}_1 \sim P}[\varphi^c(\mathbf{x}_1)] + \mathbb{E}_{\mathbf{x}_2 \sim Q}[\varphi(\mathbf{x}_2)],$$

This article has been accepted for publication in IEEE Transactions on Pattern Analysis and Machine Intelligence. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPAMI.2024.3489030

where φ^c is called weak $c-$transform,

$$\varphi^c(\mathbf{x}_1) = \inf_{P \in \mathbb{P}(\mathbb{R}^d)} c(\mathbf{x}_1, P) - \mathop{\mathbb{E}}_{\mathbf{x}_2 \sim P}[\varphi(\mathbf{x}_2)]. \tag{27}$$

Based on the weak-OT formulation, [47] proposes a max-min reformulation using convex analysis results [65] for the dual problem in eq. 26. The authors introduce a mapping $T : \mathbb{R}^d \times \mathcal{Z} \rightarrow \mathbb{R}^d$, and a distribution S (e.g., $S = \mathcal{N}(0, 1)$) for parametrizing P in eq. 27, i.e., $P = T(\mathbf{x}, \cdot)_\sharp S$. Hence,

$$\inf_{f,T} \mathop{\mathbb{E}}_{\mathbf{x}\sim Q}[\varphi(\mathbf{x})] + \mathop{\mathbb{E}}_{\mathbf{x}\sim P}[c(\mathbf{x}, T(\mathbf{x}, \cdot)_\sharp S) - \mathop{\mathbb{E}}_{z\sim S}[\varphi(T(\mathbf{x}, z))]]. \tag{28}$$

The intuition behind the mapping T is as follows. Mappings of the kind $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ are called deterministic, i.e., they map $\mathbf{x}_1 \sim P$ into $\mathbf{x}_2 \sim Q$. As we discussed in sections 2 and 3, such a mapping may not exist. As a result, mappings $T(\mathbf{x}_1, z)$ are called *stochastic*, as $\mathbf{x}_2 = T(\mathbf{x}_1, z)$ depends on $z \sim S$. In practice, [47] proposes to parametrize T and φ by NNs with parameters θ and ξ. The optimization in eq. 28 is then carried out by sampling mini-batches from P, Q and S respectively.

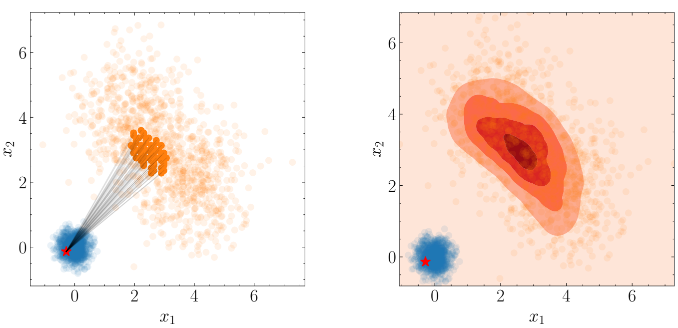

Fig. 8: Comparison between strong (a) and weak (b) OT. In (a), a point in distribution P is matched, through an OT plan, to points in Q. In (b), for $\mathbf{x}_1$ in P (red star), one has a distribution $\gamma(\cdot|\mathbf{x}_1)$ over points in Q.

4 Supervised Learning

In this section we review applications of OT for supervised learning in two directions. First, in section 4.1, we review the use of OT as a loss in classification. Second, in section 4.2, we review OT for fairness.

4.1 OT as a loss

Empirical Risk Minimization. Let $\mathcal{X} = \mathbb{R}^d$ be a feature space, and $\mathcal{Y} = \{1, \cdots, n_c\}$ be a label space. For a distribution $P \in \mathbb{P}(\mathcal{X})$, a ground truth $h_0 : \mathcal{X} \rightarrow \mathcal{Y}$, and a loss function $\mathcal{L} : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$, the risk of $h : \mathcal{X} \rightarrow \mathcal{Y}$ is,

$$\mathcal{R}_P(h) = \mathop{\mathbb{E}}_{\mathbf{x}\sim P}[\mathcal{L}(h(\mathbf{x}), h_0(\mathbf{x}))]. \tag{29}$$

In classification, one defines a family of functions $\mathcal{H} \subset \mathcal{X}^{\mathcal{Y}}$, where one searches for $h^\star \in \mathcal{H}$ that minimizes eq. 29. In practice, one does not have access to P, but rather to samples $\mathbf{x}_i^{(P)} \sim P$ and $y_i^{(P)} = h_0(\mathbf{x}_i^{(P)})$, which leads to,

$$\hat{\mathcal{R}}_P(h) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(h(\mathbf{x}_i^{(P)}), y_i^{(P)}). \tag{30}$$

In analogy to $h^\star$, one can minimize the empirical risk over $h \in \mathcal{H}$. This strategy is known as Empirical Risk Minimization (ERM) [66]. In the context of NNs, $h = h_\theta$ is parametrized by the weights θ of the network. An usual choice in ML is $\mathcal{L}(\hat{y}_i, y_i) = \sum_{k=1}^{n_c} y_{ik} \log \hat{y}_{ik}$ where $y_{ik} = 1$ if and only if the sample i belongs to class k, and $\hat{y}_{ik} \in [0, 1]$ is the predicted probability of sample i belonging to k. This choice of loss is related to Maximum Likelihood Estimation (MLE), and hence to the KL divergence (see [67, Chapter 4]) between $P(Y|X)$ and $P_\theta(Y|X)$.

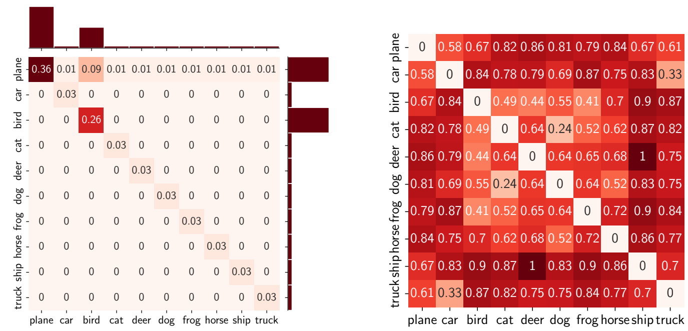

Fig. 9: (a) OT plan between two probability vectors over classes in CIFAR10 [68]. (b) Pairwise distances between Word2Vec [69] embeddings of class labels. The Wasserstein loss proposed by [70] corresponds to the Frobenius inner product between those two matrices.

As remarked by [70], this choice disregards similarities in the label space. As such, a prediction of semantically close classes (e.g., a car and a truck) yields the same loss as semantically distant ones (e.g., a dog and a ship). To remedy this issue, [70] proposes using the Wasserstein distance between the network's predictions and the ground-truth vector, $\mathcal{L}(\hat{y}, y) = \min_{\gamma \in U(\hat{y}, y)} \langle \gamma, \mathbf{C} \rangle_F$, where $\mathbf{C} \in \mathbb{R}^{n_c \times n_c}$ is a metric between labels. This is illustrated in Figure 9, in the context of the CIFAR10 dataset [68].

With respect [70], similar principles were applied by [71], [72] in the context of semantic segmentation, in which the authors use a pre-defined ground-metric corresponding to the severity of wrong classification. Likewise [73] employed a Wasserstein term between probability vectors as a regularizer term in the context of learning under noisy labels.

Distributionally Robust Optimization. Besides its contributions to the ERM framework, OT can be used as a tool in Distributionally Robust Optimization (DRO) [74], [75]. DRO can be used as a generalization of the ERM when data is noisy. The idea consists on regularizing the problem by robustifying it,

$$\hat{\mathcal{R}}_{P,\epsilon}(h) = \sup_{Q \in \mathbb{B}(\epsilon, 1, \hat{P})} \mathop{\mathbb{E}}_{\mathbf{x}\sim Q}[\mathcal{L}(h(\mathbf{x}), h_0(\mathbf{x}))], \tag{31}$$

where $\mathbb{B}_{\epsilon,1}(\hat{P})$ is the $1-$Wasserstein ball with radius ϵ around $\hat{P}$. As shown in [76], minimizing $\hat{\mathcal{R}}_{P,\epsilon}$ over $h \in \mathcal{H}$ is equivalent to a regularized ERM problem,

$$\hat{h}_{WDRO} = \inf_{h \in \mathcal{H}} \hat{\mathcal{R}}_P(h) + \epsilon \mathrm{Lip}(\mathcal{L}) \|h\|_*,$$

where $\mathrm{Lip}(\mathcal{L})$ is the Lipschitz constant of the loss $\mathcal{L}$ and $\|h\|_* = \sup\{|h(\mathbf{x})| : \mathbf{x} \in \mathcal{X}\}$ is the dual norm of h.

This article has been accepted for publication in IEEE Transactions on Pattern Analysis and Machine Intelligence. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPAMI.2024.3489030

4.2 Fairness

The analysis of biases in ML algorithms has received increasing attention, due to the now widespread use of these systems for automatizing decision-making processes. In this context, fairness [77] is a sub-field of ML concerned with achieving fair treatment to individuals in a population. Throughout this section, we focus on fairness in binary classification.

There are mainly three approaches in the fairness literature [78]: pre, in and post-processing. First, pre-processing corresponds to transforming the input data so that models trained on it achieve some notion of fairness. Second, in-processing incorporates a metric of fairness in the learning process of a model. Finally, post-processing correspond to applying transformations to a model's outputs, or performing model selection. In this section, we focus on the first two strategies. In this context OT is used to enforce statistical parity [79], which is expressed in probabilistic terms as,

$$P(h(X)=1|S=0)=P(h(X)=1|S=1) \tag{32}$$

Pre-processing. Based on equation 32, [80] proposes a pre-processing technique, based on OT, to enforce statistical parity. Let $P_s = P(X|S=s)$, $s=0,1$. Their idea is to devise a transformation T with $T_\sharp P_0 = T_\sharp P_1$. In addition, T should not destroy information, i.e. P_s and $T_\sharp P_s$ should be as close as possible. As a result, [80] chooses to realize both of these constraints with the Wasserstein barycenter $\hat{P}_t = \mathcal{B}([1-t,t];\{\hat{P}_0,\hat{P}_1\})$ (see eq. 10),

$$\hat{P}_t(\mathbf{x}) = \sum_{i=1}^{n_0}\sum_{j=1}^{n_1}\gamma_{ij}\delta(\mathbf{x}-\pi_t(\mathbf{x}_i^{(P_0)},\mathbf{x}_j^{(P_1)})),$$
$$\pi_t(\mathbf{x}_i^{(P_0)},\mathbf{x}_j^{(P_1)}) = (1-t)\mathbf{x}_i^{(P_0)} + t\mathbf{x}_j^{(P_1)}.$$

In [80], the authors use $t = 1/2$, so as to avoid disparate impact. Furthermore, trying to build T s.t. $T_\sharp P_0 = T_\sharp P_1$ may compromise too much accuracy. As a result, [80] introduce the concept of *random repair*, which corresponds to drawing $b_1,\cdots,b_{n_0+n_1} \sim Be(\lambda)$, where $Be(\lambda)$ is a Bernoulli distribution with parameter λ. For P_0 (resp. P_1),

$$\mathbf{X}^{(P_0)} = \bigcup_{i=1}^{n_0}\begin{cases}\{\mathbf{x}_i^{(P_0)}\} & \text{if } b_i = 0\\ \{\mathbf{x}_{t,ij} : \gamma_{ij} > 0\} & \text{if } b_i = 1\end{cases},$$

where the *amount of repair* λ regulates the trade-off between fairness ($\lambda=1$) and classification performance ($\lambda=0$).

In-Processing. In another direction, one may use OT for devising a regularization term to enforce fairness in ERM. Two works follow this strategy, namely [81] and [82]. In this sense, let $f = h\circ g$, be the composition of a feature extractor g and a classifier h. Therefore [81] proposes minimizing,

$$(h^\star,g^\star) = \underset{h,g}{\arg\min}\,\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}(h(g(\mathbf{x}_i)),y_i) + \beta d(P_0,P_1),$$

for $\beta > 0$, $P_0 = P(X|S=0)$ (resp $S=1$) and d being either the MMD or the Sinkhorn divergence (see section 2). On another direction, [82] considers the *output distribution* $P_s = P(h(g(X))|S=s)$. Their approach is more general, as they suppose that attributes can take more than 2 values, namely $s\in\{1,\cdots,N_S\}$. Their insight is that the output distribution should be transported to the distribution closest to each group's conditional, namely $P_k = P(h(g(X))|S=k)$. This corresponds to $P_{\bar{s}} = \mathcal{B}(\alpha;\{P_k\}_{k=1}^{N_S})$, for $\alpha_k = n_k/n$ and $n=\sum_k n_k$. As in [81], the authors enforce this condition through regularization, namely,

$$\min_{h,g}\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}(h(g(\mathbf{x}_i)),y) + \beta\sum_{k=1}^{N_S}\frac{n_k}{n}W_2(P_k,\bar{P}).$$

5 Unsupervised Learning

In this section, we discuss unsupervised learning techniques that leverage the Wasserstein distance as a loss function. We consider three cases: generative modeling (section 5.1), representation learning (section 5.2) and clustering (section 5.3).

5.1 Generative Modeling

There are mainly three types of generative models that benefit from OT, namely, GANs (section 5.1.1), Variational Autoencoders (VAEs) (section 5.1.2) and normalizing flows (section 5.1.3). As discussed in [83], [84], OT provides an unifying framework for these principles through the Minimum Kantorovich Estimator (MKE) framework, introduced by [85], which, given samples from P_0, tries to find θ that minimizes $W_2(P_\theta,P_0)$. In the following we denote by $\mathcal{X}$ the input space (e.g., image space) and $\mathcal{Z}$ to the latent space (e.g., an Euclidean space $\mathbb{R}^p$).

5.1.1 Generative Adversarial Networks

GAN is a NN architecture proposed by [86] for sampling from a complex probability distribution P_0. This architecture has two networks. First, a generator $g_\theta:\mathcal{Z}\rightarrow\mathcal{X}$, that maps $\mathbf{z}\sim Q$ to a sample $\mathbf{x}\in\mathcal{X}$. Q is assumed to be a simple distribution (e.g., $\mathcal{N}(\mathbf{0},\sigma^2\mathbf{I})$). Second, a discriminator h_ξ : $\mathcal{X}\rightarrow[0,1]$. The GAN architecture is shown in Figure 10 (a), and is trained with the following optimization problem,

$$\min_\theta\max_\xi\underset{\mathbf{x}\sim P_0}{\mathbb{E}}[\log(h_\xi(\mathbf{x}))] + \underset{\mathbf{z}\sim Q}{\mathbb{E}}[\log(1-h_\xi(g_\theta(\mathbf{z})))].$$

In other words, h_ξ is trained to maximize the probability of assigning the correct label to $\mathbf{x}$ (labeled 1) and $g_\theta(\mathbf{z})$ (labeled 0). Conversely, g_θ is trained to minimize $\log(1-h_\xi(g_\theta(\mathbf{z})))$. As a consequence it minimizes the probability that h_ξ guesses its samples correctly.

As shown in [86], for an optimal discriminator $h_{\xi^\star}$, the generator cost is equivalent to the so-called Jensen-Shannon (JS) divergence. In this sense, [44] proposed the Wasserstein GAN (WGAN) algorithm, which substitutes the JS divergence by the KR metric (equation 6),

$$\min_\theta\max_{h_\xi\in\text{Lip}_1}\mathcal{L}_W(\theta,\xi) = \underset{\mathbf{x}\sim P_0}{\mathbb{E}}\Big[h_\xi(\mathbf{x})\Big] - \underset{\mathbf{z}\sim Q}{\mathbb{E}}\Big[h_\xi(g_\theta(\mathbf{z}))\Big].$$

Nonetheless, the WGAN involves maximizing $\mathcal{L}_W$ over $h_\xi\in\text{Lip}_1$, which is not straightforward for NNs. Possible solutions are: (i) clipping the weights ξ [44]; (ii) penalizing the gradients of h_ξ [87]; (iii) normalizing the spectrum of the weight matrices [88]. Surprisingly, even though (ii) and (iii) improve over (i), several works have confirmed that the WGAN and its variants *do not estimate the Wasserstein distance* [89], [90], [91].

In another direction, one can calculate $\nabla_\theta W_p$ through the primal problem (eq. 14). To alleviate the computational

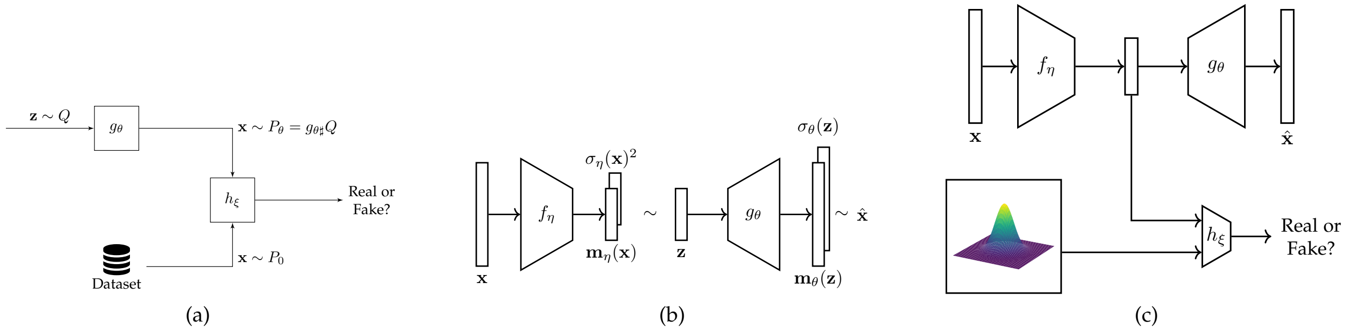

Fig. 10: Different architectures in generative modeling. (a) shows GANs, which is composed of a generator of synthetic samples, and a discriminator of real and fake samples. (b) shows VAEs, which are a stochastic version of auto-encoders, an architecture composed of an encoder and a decoder. (c) shows the architecture of Adversarial Autoencoders (AAEs), which combine ideas of GANs and VAEs.

and memory complexity of OT, one uses mini-batches [26], [53] and computes the Sinkhorn divergence [26]. It remains the question on how to calculate the gradients w.r.t. θ. First, [26] proposes to back-propagate the gradients through the Sinkhorn iterations, but it is commonly time consuming and numerically unstable. Second [92] and [93] advocate to take the gradients at convergence of Sinkhorn iterations, i.e. assuming that the transport plan does not depend on θ. This is justified by the Envelope theorem [94].

In addition, as discussed in [26], an Euclidean ground-cost is likely not meaningful for complex data (e.g., images). The authors thus propose learning a parametric ground cost $(\mathbf{C}_\eta)_{ij} = \|f_\eta(\mathbf{x}_i^{(P_\theta)}) - f_\eta(\mathbf{x}_j^{(P_0)})\|_2^2$, where $f_\eta : \mathcal{X} \rightarrow \mathcal{Z}$ is a NN that learns a representation for $\mathbf{x} \in \mathcal{X}$. Overall the optimization problem proposed by [26] is,

$$\min_\theta \max_\eta S_{c_\eta,\epsilon}(P_\theta, P_0). \tag{33}$$

We stress the fact that *engineering/learning* the ground-cost c_η is important for having a meaningful metric between distributions, since it serves to compute distances between samples. For instance, the Euclidean distance is known to not correlate well with perceptual or semantic similarity between images [95].

Finally, as in [32], [33], [96], [97], one can employ sliced Wasserstein metrics (see section 3.1). This has two advantages: (i) computing the sliced Wasserstein distances is computationally less complex, (ii) these distances are more robust w.r.t. the curse of dimensionality [98], [99]. These properties favour their usage in generative modeling, as data is commonly high dimensional.

5.1.2 Autoencoders

In a parallel direction, one can leverage autoencoders for generative modeling. This idea was introduced in [100], who proposed using stochastic encoding and decoding functions. Let $f_\eta : \mathcal{X} \rightarrow \mathcal{Z}$ be an encoder network. Instead of mapping $\mathbf{z} = f_\eta(\mathbf{x})$ deterministically, f_η predicts a mean $\mathbf{m}_\eta(\mathbf{x})$ and a variance $\sigma_\eta(\mathbf{x})^2$ from which the code $\mathbf{z}$ is sampled, that is, $\mathbf{z} \sim Q_\eta(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{m}_\eta(\mathbf{x}), \sigma_\eta(\mathbf{x})^2)$. The stochastic decoding function $g_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ works similarly for reconstructing the input $\hat{\mathbf{x}}$. This is shown in figure 10 (b). In this framework, the decoder plays the role of generator.

The VAE framework is built upon variational inference, which is a method for approximating probability densities [101]. Indeed, for a parameter vector θ, generating new samples $\mathbf{x}$ is done in two steps: (i) sample $\mathbf{z}$ from the prior $Q(\mathbf{z})$ (e.g., a Gaussian), then (ii) sample $\mathbf{x}$ from the conditional $P_\theta(\mathbf{x}|\mathbf{z})$. The issue comes from calculating the marginal,

$$P_\theta(\mathbf{x}) = \int Q(\mathbf{z}) P_\theta(\mathbf{x}|\mathbf{z}) d\mathbf{z},$$

which is intractable. This hinders the MLE, which relies on $\log P_\theta(\mathbf{x})$. VAEs tackle this problem by first considering an approximation $Q_\eta(\mathbf{z}|\mathbf{x}) \approx P_\theta(\mathbf{z}|\mathbf{x})$. Secondly, one uses the Evidence Lower Bound (ELBO),

$$\text{ELBO}(Q_\eta) = \mathop{\mathbb{E}}_{\mathbf{z}\sim Q_\eta}[\log P_\theta(\mathbf{x}, \mathbf{z})] - \mathop{\mathbb{E}}_{\mathbf{z}\sim Q_\eta}[\log Q_\eta(\mathbf{z})],$$

instead of the log-likelihood. Indeed, as shown in [100], the ELBO is a lower bound for the log-likelihood. As follows, one turns to the maximization of the ELBO, which can be rewritten as,

$$\mathcal{L}(\theta, \eta) = \mathop{\mathbb{E}}_{\mathbf{x}\sim P_0}\left[\mathop{\mathbb{E}}_{\mathbf{z}\sim Q_\eta}[\log P_\theta(\mathbf{x}|\mathbf{z})] - \text{KL}(Q_\eta \| Q)\right]. \tag{34}$$

A somewhat middle ground between the frameworks of [86] and [100] is the AAE architecture [102], shown in figure 10 (c). AAEs are different from GANs in two points, (i) an encoder f_η is added, for mapping $\mathbf{x} \in \mathcal{X}$ into $\mathbf{z} \in \mathcal{Z}$; (ii) the adversarial component is done in the latent space $\mathcal{Z}$. While the first point puts the AAE framework conceptually near VAEs, the second shows similarities with GANs.

Based on both AAEs and VAEs, [103] proposed the Wasserstein Autoencoder (WAE) framework, which is a generalization of AAEs. Their insight is that, when using a deterministic decoding function g_θ, one may simplify the MK formulation,

$$\inf_{\gamma\in\Gamma} \mathop{\mathbb{E}}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}[c(\mathbf{x}_1, \mathbf{x}_2)] = \inf_{Q_\eta = Q} \mathop{\mathbb{E}}_{\mathbf{x}\sim P_0}\left[\mathop{\mathbb{E}}_{\mathbf{z}\sim Q_\eta}[c(\mathbf{x}, g_\theta(\mathbf{z}))]\right].$$

As follows, [103] suggests enforcing the constraint in the infimum through a penalty term Ω, leading to,

$$\mathcal{L}(\theta, \eta) = \mathop{\mathbb{E}}_{\mathbf{x}\sim P_0}\left[\mathop{\mathbb{E}}_{\mathbf{z}\sim Q_\eta}[c(\mathbf{x}, g_\theta(\mathbf{z}))]\right] + \lambda\Omega(Q_\eta, Q),$$

for $\lambda > 0$. This expression is minimized jointly w.r.t. θ and η. As choices for the penalty Ω, [103] proposes using either the JS divergence, or the MMD distance. In the first case the formulation falls back to a mini-max problem similar to the AAE framework. A third choice was proposed by [104], which relies on the Sinkhorn loss $S_{c,\epsilon}$, thus leveraging the work of [26].

5.1.3 Continuous Normalizing Flows

OT is linked to Normalizing Flows (NFs), especially Continuous NFs (CNFs), through its dynamical formulation. We thus focus on this class of algorithms. For a broader review, we refer the readers to [105]. NFs rely on the chain rule for computing changes in probability distributions. Let $\mathbf{T} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a diffeomorphism. If, for $\mathbf{z} \sim Q$ and $\mathbf{x} \sim P$, $\mathbf{z} = \mathbf{T}(\mathbf{x})$, one has,

$$\log P(\mathbf{x}) = \log Q(\mathbf{z}) - \log|\det \nabla \mathbf{T}^{-1}|.$$

Following [105], a NF is characterized by a sequence of diffeomorphisms $\{T_i\}_{i=1}^N$ that transform a simple probability distribution (e.g., a Gaussian distribution) into a complex one. If one understands this sequence *continuously*, they can be modeled through dynamic systems. In this context, CNFs are a class of generative models based on dynamic systems,

$$\mathbf{z}(0, \mathbf{x}) = \mathbf{x}, \text{ and, } \dot{\mathbf{z}}(t, \mathbf{x}) = \mathbf{F}_\theta(\mathbf{z}(t, \mathbf{x})). \tag{35}$$

which is an Ordinary Differential Equation (ODE). Let $\mathbf{z}_t(\mathbf{x}) = \mathbf{z}(t, \mathbf{x})$. By discretizing the time variable, equation 35 becomes $\mathbf{z}_{n+1} = \mathbf{z}_n + \tau \mathbf{F}_\theta(\mathbf{z}_n)$, for a step-size $\tau > 0$. This equation is similar to the ResNet structure [106]. Indeed, this is the insight behind the seminal work of [107], who proposed the Neural ODE (NODE) algorithm for parametrizing ODEs through NNs. CNFs are thus the intersection of NFs with NODE. The advantage of the approach of [107] is that, associated with eq. 35, the log-probability follows an ODE as well,

$$\frac{\partial}{\partial t} \log|\det \nabla \mathbf{z}| = \mathrm{Tr}(\nabla_{\mathbf{z}} \mathbf{F}_\theta) = \mathrm{div}(\mathbf{F}_\theta). \tag{36}$$

With this workaround, minimizing the negative log-likelihood amounts to minimizing,

$$\sum_{i=1}^n - \log Q(\mathbf{z}_T(\mathbf{x}_i)) - \int_0^T \mathrm{div}(\mathbf{F}_\theta(\mathbf{z}_t(\mathbf{x}_i)))dt, \tag{37}$$

which can be solved using the automatic differentiation [107]. One limitation of CNFs is that the generic flow can be highly irregular, having unnecessarily fluctuating dynamics. As remarked by [108], this can pose difficulties to numerical integration. These issues motivated the proposition of two regularization terms for simpler dynamics. The first term is based dynamic OT. Let,

$$\rho_t = \mathbf{z}_{t,\sharp} P = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{z}_t(\mathbf{x}_i)},$$

which simplifies eq. 7 to,

$$\begin{aligned} \min_\theta \quad & \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \int_0^T \|\mathbf{F}_\theta(\mathbf{z}_t(\mathbf{x}_i))\|^2 dt, \\ \text{subject to} \quad & \mathbf{z}_0 = \mathbf{x}, \text{ and } \mathbf{z}_{T,\sharp} P = Q, \end{aligned} \tag{38}$$

where $\|\mathbf{F}_\theta(\mathbf{z}_t(\mathbf{x}_i))\|^2$ is the kinetic energy of the particle $\mathbf{x}_i$ at time t, and $\mathcal{L}$ to the whole kinetic energy. In addition, ρ_t has the properties of an OT map. Among these, OT enforces that: (i) the flow trajectories do not intersect, and (ii) the particles follow geodesics w.r.t. the ground cost (e.g., straight lines for an Euclidean cost), thus enforcing the regularity of the flow. Nonetheless, as [108] remarks, these properties are enforced only on training data. To effectively generalize them, the authors propose a second regularization term, that consists on the Frobenius norm of the Jacobian,

$$\Omega(\mathbf{F}) = \frac{1}{n} \sum_{i=1}^n \|\nabla \mathbf{F}_\theta(\mathbf{z}_T(\mathbf{x}_i))\|_F^2. \tag{39}$$

Thus, the Regularized Neural ODE (RNODE) algorithm combines equation 36 with the penalties in equations 38 and 39. Ultimately, this is equivalent to minimizing the KL divergence $\mathrm{KL}(\mathbf{z}_{T,\sharp} P \| Q)$ with the said constraints.

5.2 Dictionary Learning

Dictionary learning [109] represents data $\mathbf{X} \in \mathbb{R}^{n \times d}$ through a set of K atoms $\mathbf{D} \in \mathbb{R}^{k \times d}$ and n representations $\mathbf{A} \in \mathbb{R}^{n \times k}$. For a loss $\mathcal{L}$ and a regularization term Ω,

$$\underset{\mathbf{D},\mathbf{A}}{\arg\min} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(\mathbf{x}_i, \mathbf{D}^T \mathbf{a}_i) + \lambda \Omega(\mathbf{D}, \mathbf{A}). \tag{40}$$

In this setting, the data points $\mathbf{x}_i$ are approximated linearly through the matrix-vector product $\mathbf{D}^T \mathbf{a}_i$. In other words, $\mathbf{X} \approx \mathbf{AD}$. The practical interest is finding a faithful and sparse representation for $\mathbf{X}$. In this sense, $\mathcal{L}$ is often the Euclidean distance, and $\Omega(\mathbf{A})$ the ℓ_1 or ℓ_0 norm of vectors.

When the elements of $\mathbf{X}$ are non-negative, the dictionary learning problem is known as Non-negative Matrix Factorization (NMF) [110]. This scenario is particularly useful when data are histograms, that is, $\mathbf{x}_i \in \Delta_d$. As such, one needs to choose an appropriate loss function for histograms. OT provides such a loss, through the Wasserstein distance [111], in which case the problem can be solved using BCD: for fixed $\mathbf{D}$, solve for $\mathbf{A}$, and vice-versa.

Nonetheless, the BCD iterations scale poorly since NMF is equivalent to solving n linear programs with d^2 variable. To circumvent this issue, [112] propose using the Sinkhorn algorithm [25] for NMF. Besides reducing complexity, using the Sinkhorn distance $W_{c,\epsilon}$ makes NMF smooth, thus gradient descent methods can be applied successfully. The optimization problem of Wasserstein NMF (WNMF) is,

$$\underset{\mathbf{D},\mathbf{A}}{\arg\min} \frac{1}{N} \sum_{i=1}^N W_{p,\epsilon}(\mathbf{x}_i, \mathbf{D}^T \mathbf{a}_i) + \lambda \Omega(\mathbf{D}, \mathbf{A}), \tag{41}$$

subject to $\mathbf{AD} \in (\Delta_d)^N$. Assuming $\mathbf{a}_i \in \Delta_k$ implies that $\mathbf{D}^T \mathbf{a}_i$ represents a weighted average of dictionary elements. This strategy can be understood as a barycenter under the Euclidean metric. Alternatively, [113] proposes to calculates barycenters on the Wasserstein space, that is,

$$\underset{\mathbf{D},\mathbf{A}}{\arg\min} \sum_{i=1}^N W_{p,\epsilon}(\mathcal{B}(\mathbf{a}_i, \mathbf{D}), \mathbf{x}_i) + \lambda \Omega(\mathbf{D}, \mathbf{A}).$$

As a result, [113] perform a non-linear aggregation of atoms.

We show a comparison of these strategies in Figure 11, for a problem in which the histograms are Gaussian distributions $\mathbf{x}_\ell = \mathcal{N}(m_\ell, 1)$, $m_\ell = (1-\ell/3)m_0 + (\ell/3)m_1$, with $m_0 = -6$, $m_1 = 6$, and $\ell = 0, \cdots, 3$. As a result, the underlying set of histograms can be conveniently expressed in a Wasserstein space (see our discussion in sec. 2), which illustrates the advantage of Wasserstein Dictionary Learning (WDL). Generally, the advantage of WDL is non-linearly interpolating atom distributions, allowing the learned dictionary to have more expressivity.

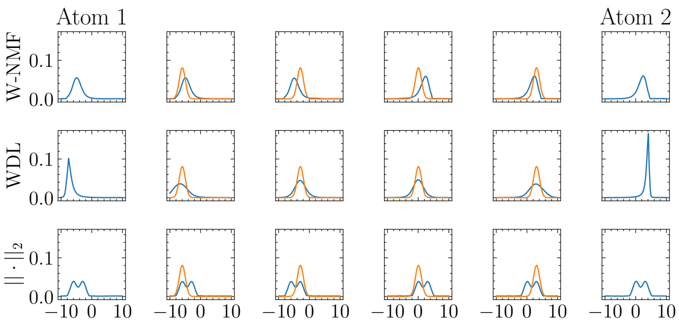

Fig. 11: Comparison of WNMF [112] (1st row), WDL [113] (2nd row) and NMF (3rd row) over histograms (1st row).

Graph Dictionary Learning. An interesting use case of this framework is Graph Dictionary Learning (GDL). Let $\{G_\ell : (\mathbf{p}_\ell, \mathbf{S}_\ell\}_{\ell=1}^N$ denote a dataset of N graphs encoded by their node similarity matrices $\mathbf{S}_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, and histograms over nodes $\mathbf{p}_\ell \in \Delta_{n_\ell}$. [114] proposes,

$$\underset{\{\mathbf{S}_k\}_{k=1}^K, \{\mathbf{a}_\ell\}_{\ell=1}^N}{\arg\min} \sum_{\ell=1}^N \text{GW}_2\left(\mathbf{S}_\ell, \sum_{k=1}^K a_{\ell,k}\mathbf{S}_k\right)^2 - \lambda\|\mathbf{a}_\ell\|_2^2,$$

in analogy to eq. 40, using the GW distance (see eq. 24) as loss function. Parallel to this line of work, [115] proposes Gromov-Wasserstein Factorization (GWF), which approximates $\mathbf{S}^{(\ell)}$ non-linearly through GW-barycenters [62].

Topic Modeling. Dictionary learning can be used for topic modeling [116], in which a set of documents $\{\hat{P}_i\}_{i=1}^N$ is expressed in terms of a dictionary of topics $\{\hat{Q}_k\}_{k=1}^K$ weighted by mixture coefficients. Indeed, documents can be interpreted as probability distributions over words in a vocabulary $V = \{w_1, \cdots, w_n\}$ [117]. In this context [118] proposes an OT-inspired algorithm for learning a set of distributions $\mathcal{Q} = \{\hat{Q}_k\}_{k=1}^K$ over words, which represent *a topic*. Learning is based on a hierarchical OT problem,

$$\min_{\{\hat{Q}_k\}_{k=1}^K, \mathbf{b}, \gamma} \sum_{k=1}^K \sum_{i=1}^N \gamma_{ik} W_{2,\epsilon}(\hat{Q}_k, \hat{P}_i) - H(\gamma),$$

where $\hat{Q}_k = \sum_{v=1}^n q_{k,v}\delta_{\mathbf{x}_v^{(Q_k)}}$, and $\mathbf{b} = [b_1, \cdots, b_K]$ is the coefficient vector of topics. In this sense $\sum_i \gamma_{ik} = b_k$, and $\sum_k \gamma_{ik} = p_i$, with $p_i = {}^{n_i}/_{n}$, i.e., the proportion of words in document $\hat{P}_i$. Note that, when topics $\mathcal{Q}$ are computed, one can compute hierarchical distances between documents P_i and P_j through the Hierarchical Optimal Transport Topic (HOTT) distance [119],

$$\text{HOTT}(\hat{P}_i, \hat{P}_j) = W_1\left(\sum_{k=1}^K d_{ik}\delta_{\hat{Q}_k}, \sum_{k=1}^K d_{jk}\delta_{\hat{Q}_k}\right),$$

where $\mathbf{d}_i, \mathbf{d}_j \in \Delta_K$ are document distributions over topics and the ground-cost is the Word Mover Distance (WMD) [117], hence the term *hierarchical*.

5.3 Clustering

Clustering is a problem within unsupervised learning that deals with the aggregation of features into groups [120]. From the perspective of OT, this is linked to the notion of quantization [121], [122]; Indeed, from a distributional viewpoint, given $\mathbf{X}^{(P)} \in \mathbb{R}^{n \times d}$, quantization corresponds to finding the matrix $\mathbf{X}^{(Q)} \in \mathbb{R}^{K \times d}$ minimizing $W_2(P, Q)$. This has been explored in a number of works, such as [123], [124] and [125]. In this sense, OT serves as a framework for quantization/clustering, thus allowing for theoretical results such as convergence bounds for the K-means algorithm [122], as well as extensions [126].

Co-Clustering. While standard clustering can be seen as a method for grouping the rows of a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, co-clustering seeks to group rows and columns simultaneously. OT contributed to this setting in 2 ways [127]. First, the Co-Clustering OT (CCOT) strategy relies on row and column distributions,

$$\hat{P} = \frac{1}{n}\sum_{i=1}^n \delta_{\mathbf{r}_i^{(P)}}, \text{ and, } \hat{Q} = \frac{1}{d}\sum_{j=1}^d \delta_{\mathbf{c}_j^{(Q)}},$$

where $\mathbf{r}_i^{(P)} = [x_{1,i}, \cdots, x_{n,i}] \in \mathbb{R}^n$ and $\mathbf{c}_j^{(Q)} = [x_{j,1}, \cdots, x_{j,d}] \in \mathbb{R}^d$. Note that, as discussed in [127, Section 3.6], OT is only defined for $n \geq d$, in which case one can sub-sample the rows of $\mathbf{X}$ for defining a $d \times d$ OT problem. Clustering over rows and columns is done by jump detection [128] on vectors $(\mathbf{f}, \mathbf{g})$ defined by the Sinkhorn algorithm (cf. eq. 16).

Second, the CCOT-GW strategy uses the Gromov extension of OT (cf. eq. 24) between kernel matrices $K_r \in \mathbb{R}^{n \times n}$, $K_c \in \mathbb{R}^{d \times d}$ and $K \in \mathbb{R}^{p \times p}$, for an hyper-parameter $p \in \mathbb{N}$. The matrix K is the GW barycenter [62] of K_r and K_c, and K_r, K_c capture the intra-rows and intra-columns similarities. Clustering is done similarly to CCOT, i.e., one solves 2 entropic regularized Gromov OT problems, between K_r and K, and between K and K_c.

Gaussian Mixture Models (GMMs) are a kind of probabilistic model defined by $\theta = \{\beta_k, \mathbf{m}_k, \mathbf{S}_k\}_{k=1}^K$, where,

$$P_\theta(\mathbf{x}) = \sum_{k=1}^K \beta_k \mathcal{N}(\mathbf{x}|\mathbf{m}_k, \mathbf{S}_k), \tag{42}$$

where $\sum \beta_k = 1$, $\beta_k \geq 0$. From the perspective of clustering, each $(\beta_k, \mathbf{m}_k, \mathbf{S}_k)$ represents a group of data points. From eq. 42 one can get soft-assignments for $\mathbf{x}$ using,

$$\alpha_k(\mathbf{x}) = \frac{\beta_k \mathcal{N}(\mathbf{x}|\mathbf{m}_k, \mathbf{S}_k)}{\sum_j \beta_j \mathcal{N}(\mathbf{x}|\mathbf{m}_j, \mathbf{S}_j)},$$

which draws a parallel between clustering and generative modeling. Learning a GMM is usually done via maximum likelihood, which ultimately amounts to minimizing $\text{KL}(P_0|P_\theta)$, where P_0 represents the data distribution In this context [96] proposed learning P_θ by minimizing $\text{SW}_p(P_0, P_\theta)$ (see sec 3.1). This problem can be nicely written in closed form, due to the properties of Gaussian distributions. Let $P = \mathcal{N}(\mathbf{m}, \mathbf{S})$, then [96] shows $\pi_{\mathbf{u},\sharp}P =$

$\mathcal{N}(\langle \mathbf{u}, \mathbf{m} \rangle, \mathbf{u}^T \mathbf{S} \mathbf{u})$. As shown in [96] SW-based GMMs are robust to parameter initialization.

6 Transfer Learning

Transfer Learning (TL) is a ML framework, concerned with learning scenarios in which data follows different probability distributions. Following [129], TL can be formalized through the notions of domain and task. In the first case, *a domain* is a pair $\mathcal{D} = (\mathcal{X}, P(X))$ of a feature space (e.g., $\mathbb{R}^d$) and a feature marginal distribution. In the second case, *a task* is a pair $\mathcal{T} = (\mathcal{Y}, P(Y|X))$ of a label space (e.g., $\{1, \cdots, n_c\}$) and a conditional distribution $P(Y|X)$. Given a source $(\mathcal{D}_S, \mathcal{T}_S)$ and a target $(\mathcal{D}_T, \mathcal{T}_T)$, the goal of TL is *improving performance on the target, given knowledge from the source*.

The *distributional shift* occurring in TL can be modeled via $P_S(X, Y) \neq P_T(X, Y)$ for a source S and a target T. Since $P(X, Y) = P(X)P(Y|X) = P(Y)P(X|Y)$, three types of shift may occur [130]: (i) *covariate shift*, that is, $P_S(X) \neq P_T(X)$, (ii) *concept shift*, namely, $P_S(Y|X) \neq P_T(Y|X)$ or $P_S(X|Y) \neq P_T(X|Y)$, and (iii) *target shift*, for which $P_S(Y) \neq P_T(Y)$. In the following, we focus on Unsupervised Domain Adaptation (UDA), a setting in which one has access to labeled data from the source domain, and unlabeled data from the target domain.

6.1 Shallow Domain Adaptation

Under covariate shift, OT can be used for matching $P_S(X)$ and $P_T(X)$. This is straightforward with the Monge problem, in which $T_\sharp P_S = P_T$, but its solution may not exist in the discrete setting (e.g., section 3). Given this shortcoming, [39] uses the barycentric mapping,

$$T_\gamma(\mathbf{x}_i^{(P_S)}) = \underset{\mathbf{x} \in \mathbb{R}^d}{\arg\min} \sum_{j=1}^{n_T} \gamma_{ij} c(\mathbf{x}, \mathbf{x}_j^{(P_T)}), \tag{43}$$

where $\gamma = \text{OT}(\mathbf{p}_S, \mathbf{p}_T, \mathbf{C})$. The case where $c(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|_2^2$ is particularly interesting, as $\mathbf{T}_\gamma$ has closed-form in terms of the support $\mathbf{X}^{(P_T)} \in \mathbb{R}^{n_T \times d}$ of $\hat{P}_T$, $\hat{\mathbf{X}}^{(P_S)} = \text{diag}(\mathbf{p}_S)^{-1} \gamma \mathbf{X}^{(P_T)}$. As follows, each point $\mathbf{x}_i^{(P_S)}$ is mapped into $\hat{\mathbf{x}}_i^{(P_S)}$, which is a convex combination of the points $\mathbf{x}_j^{(P_T)}$ that receives mass from $\mathbf{x}_i^{(P_S)}$, namely, $\gamma_{ij} > 0$. This effectively generates a new training dataset $\{(\hat{\mathbf{x}}_i^{(P_S)}, y_i^{(P_S)})\}_{i=1}^{n_S}$, which hopefully leads to a classifier $\hat{h}$ that works well on $\mathcal{D}_T$. An illustration is shown in Figure 12.

The barycentric mapping has two limitations. First, it may map points to the wrong side of the decision boundary. As noted by [39], this happens when γ_{ij} moves mass between different classes. To avoid that, [39] proposes to further regularize OT plans:

$$\gamma^\star = \underset{\gamma \in U(\mathbf{p}_S, \mathbf{p}_T)}{\arg\min} \langle \mathbf{C}, \gamma \rangle_F - \epsilon H(\gamma) + \eta \Omega(\gamma; \mathbf{y}^{(P_S)}, \mathbf{y}^{(P_T)}),$$

where Ω is a class-based regularizer. This additional regularization enforces that $\gamma_{ij} > 0$ if and only if $\mathbf{x}_i^{(P_S)}$ and $\mathbf{x}_j^{(P_T)}$ have the same class. We highlight the wrong pairings done by OT in Figs. 12 and 13. The regularization strategies are divided into semi-supervised and unsupervised. In the former case, one uses a few labeled samples in the target domain to penalize $\{\gamma_{ij} > 0, y_i^{(P_S)} \neq y_j^{(P_T)}\}$. In the latter case, one penalizes transport plans such that $\mathbf{x}_j^{(P_T)}$ receives mass from $\mathbf{x}_i^{(P_S)}$ with different classes. This is achieved through group-sparsity [39, Eq. 17].

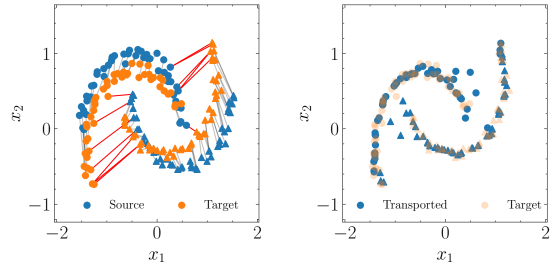

Fig. 12: Optimal Transport for Domain Adaptation (OTDA) methodology: (a) Source (red) and target (blue) samples follow different distributions; A classifier (dashed black line) fit on source is not adequate for target; (b) T_γ then transports source domain samples to the target domain.

Second, through eq. 43, T_γ it is only defined on the samples $\mathbf{x}_i^{(P_S)}$ in $\mathbf{X}^{(P_S)}$. It is thus undefined for new samples coming from P_S. There are three strategies to solve this issue. First, [38] proposes to define T_γ for new points $\mathbf{x}^{(P_S)}$ through its nearest neighbors. This, however, introduces quantization effects on the mapping. This issue motivates [131] to parametrize T_γ as a linear, or a kernel mapping. Finally, [45] proposes a large scale strategy in which the barycentric mapping is parametrized through a neural net (see section 3.3).

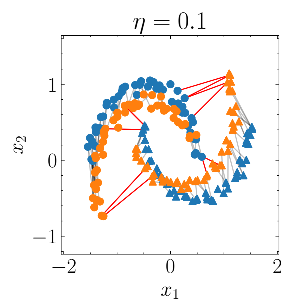

Fig. 13: Optimal transport with class regularization induces a match between $\hat{P}_S$ and $\hat{P}_T$ that tends to be more class sparse.

In a more general direction, one may assume that *the shift occurs at the level of joint distributions*. In UDA this is not directly possible, since the labels from $P_T(X, Y)$ are not available. A workaround was introduced in [132], where the authors propose a proxy distribution $P_T^h(X, h(X))$:

$$\hat{P}_T^h = \frac{1}{n_T} \sum_{j=1}^{n_T} \delta_{\mathbf{x}_j^{(P_T)}, h(\mathbf{x}_j^{(P_T)})}.$$

where $h \in \mathcal{H}$ is a classifier. As follows, [132] propose minimizing the Wasserstein distance $W_c(\hat{P}_S, \hat{P}_T^h)$ over possible classifiers $h \in \mathcal{H}$. This yields a joint minimization problem,

$$\min_{\substack{h \in \mathcal{H} \\ \gamma \in U(\mathbf{p}_S, \mathbf{p}_T)}} \sum_{i=1}^{n_S} \sum_{j=1}^{n_T} c(\mathbf{x}_i^{(P_S)}, y_i^{(P_S)}, \mathbf{x}_j^{(P_T)}) \gamma_{ij} + \lambda \Omega(h).$$

The cost c can be designed in terms of two factors, a feature loss $c_f(\mathbf{x}_i^{(P_S)}, \mathbf{x}_j^{(P_T)})$ (e.g., the Euclidean distance), and a label loss $c_\ell(y_i^{(P_S)}, h(\mathbf{x}_j^{(P_T)}))$ (e.g., the Hinge loss). As proposed in [132], the importance of c_f and c_ℓ can be controlled by a parameter α,

$$C_{ij} = \alpha c_f(\mathbf{x}_i^{(P_S)}, \mathbf{x}_j^{(P_T)}) + c_\ell(y_i^{(P_S)}, h(\mathbf{x}_j^{(P_T)})).$$

This article has been accepted for publication in IEEE Transactions on Pattern Analysis and Machine Intelligence. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TPAMI.2024.3489030

6.2 Deep Domain Adaptation

In this section, we assume that a deep NN is a composition $f = h_\xi \circ g_\theta$ of a feature extractor g with parameters θ, and a classifier h_ξ with parameters ξ. The intuition behind deep Domain Adaptation (DA) is to force the feature extractor to *learn domain invariant features*. Given $\mathbf{x}_i^{(P_S)} \sim P_S$ and $\mathbf{x}_j^{(P_T)} \sim P_T$, an invariant feature extractor g_θ should suffice,

$$(g_\theta)_\sharp \hat{P}_S = \frac{1}{n_S}\sum_{i=1}^{n_S}\delta_{\mathbf{z}_i^{(P_S)}} \overset{\mathcal{D}}{\approx} \frac{1}{n_T}\sum_{j=1}^{n_T}\delta_{\mathbf{z}_j^{(P_T)}} = (g_\theta)_\sharp \hat{P}_T,$$

where $\mathbf{z}_i^{(P_S)} = g_\theta(\mathbf{x}_i^{(P_S)})$ (resp. $\hat{P}_T$), and $\hat{P}_S \overset{\mathcal{D}}{\approx} \hat{P}_T$ means $\mathcal{D}(\hat{P}_S, \hat{P}_T) \approx 0$ for a given divergence or distance between distributions. This implies that, after the application of g_θ, distributions $\hat{P}_S$ and $\hat{P}_T$ are close. In this sense, this condition is enforced by adding an additional term to the classification loss function (e.g., the MMD as in [133]). In this context, [134] proposes the Domain Adversarial Neural Network (DANN) algorithm, based on the loss function,

$$\mathcal{L}_{\text{DANN}}(\theta, \xi, \eta) = \hat{\mathcal{R}}_{P_S}(h_\xi \circ g_\theta) - \lambda \mathcal{L}_{\mathcal{H}}(\theta, \eta),$$

where $\mathcal{R}_{P_S}$ denotes the empirical risk (see eq. 30) and $\mathcal{L}_{\mathcal{H}}$,

$$\frac{1}{n_S}\sum_{i=1}^{n_S}\log h_\eta(\mathbf{z}_i^{(P_S)})) + \frac{1}{n_T}\sum_{j=1}^{n_T}\log(1 - h_\eta(\mathbf{z}_j^{(P_T)})),$$

where h_η is a supplementary classifier, that *discriminates* between source (labeled 0) and target (labeled 1). The DANN algorithm is a mini-max optimization problem,

$$\min_{\theta,\xi}\max_{\eta} \mathcal{L}_{\text{DANN}}(\theta, \xi, \eta),$$

so as to minimize classification error and *maximize domain confusion*. This draws a parallel with the GAN algorithm of [86], presented in section 5.1.1. This remark motivated [135] for using the Wasserstein distance instead of $\mathcal{L}_{\mathcal{H}}$. Their algorithm is called Wasserstein Distance Guided Representation Learning (WDGRL), which uses as loss:

$$\mathcal{L}_W(\theta, \eta) = \frac{1}{n_S}\sum_{i=1}^{n_S} h_\eta(g_\theta(\mathbf{x}_i^{(P_S)})) - \frac{1}{n_T}\sum_{j=1}^{n_T} h_\eta(g_\theta(\mathbf{x}_j^{(P_T)})).$$

Following our discussion on KR duality (see section 2) as well as the Wasserstein GAN (see section 5.1.1), one needs to maximize $\mathcal{L}_W$ over $h_\eta \in \text{Lip}_1$. [135] proposes doing so using the gradient penalty term of [87],

$$\min_{\theta}\min_{\xi} \hat{\mathcal{R}}_{P_S}(h_\xi \circ g_\theta) + \lambda_2 \max_{\eta} \mathcal{L}_W(\theta, \eta) - \lambda_1 \Omega(h_\eta),$$

where λ_1 controls the gradient penalty term, and λ_2 controls the importance of the domain loss term.

Finally, we highlight that the Joint Distribution Optimal Transport (JDOT) framework can be extended to deep DA. First, one includes the feature extractor g_θ in the ground-cost. For $\mathbf{z}_i^{(P_S)} = g_\theta(\mathbf{x}_i^{(P_S)})$ (resp. P_T),

$$C_{ij}(\theta, \xi) = \alpha\|\mathbf{z}_i^{(P_S)} - \mathbf{z}_j^{(P_T)}\|^2 + c_\ell(y_i^{(P_S)}, h_\xi(\mathbf{z}_j^{(P_T)})),$$

second, the objective function includes a classification loss $\mathcal{R}_{P_S}$, and the OT loss, that is,

$$\mathcal{L}_{JDOT}(\theta, \xi) = \hat{\mathcal{R}}_{P_S}(h_\xi \circ g_\theta) + \sum_{i=1}^{n_S}\sum_{j=1}^{n_T}\gamma_{ij}C_{ij}(\theta, \xi), \quad (44)$$

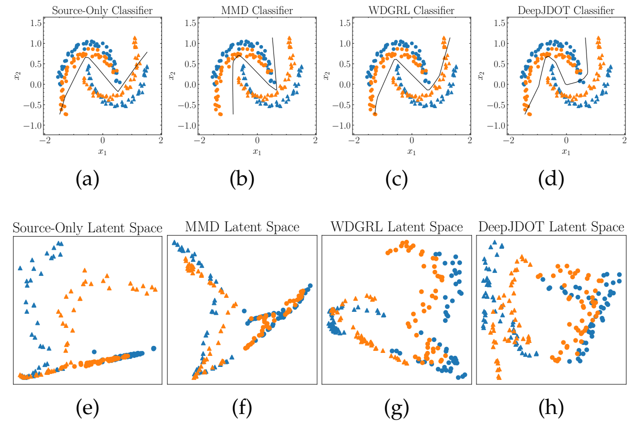

Fig. 14: Comparison of Deep DA strategies, based on the MMD (b, f) and OT (c, d, g, h). Overall, (e – h) show the PCA (2 components) of the latent space of a neural net. Note that deep DA match $\hat{P}_S$ and $\hat{P}_T$ in such space. As a result, they are able to find a classifier that correctly predicts on target domain samples.

which is jointly minimized w.r.t. $\gamma \in \mathbb{R}^{n_S \times n_T}$, θ and ξ. [51] proposes minimizing $\mathcal{L}_{\text{JDOT}}$ jointly w.r.t. θ and ξ, using mini-batches (see section 3.4). Nonetheless, the authors noted that one needs large mini-batches for a stable training. To circumvent this issue [54] proposed using *unbalanced OT* (see sec. 3.5), allowing for smaller batch sizes and improved performance.

6.3 Extensions to Domain Adaptation

Multi-Source Domain Adaptation (MSDA) considers the problem of DA when source data comes from multiple, distributionally heterogeneous domains, namely $P_{S_1}, \cdots, P_{S_K}$. In this sense, [136] proposes using the Wasserstein barycenter for building an intermediate domain $\hat{P}_B = \mathcal{B}(\alpha; \{\hat{P}_{S_k}\}_{k=1}^{N_S})$ (see equation 10), for $\alpha_k = {}^1\!/_{N_S}$. Since $\hat{P}_B \neq \hat{P}_T$, the authors propose using an additional adaptation step for transporting the barycenter towards the target. Furthermore, [137] proposes the Weighted JDOT algorithm, which generalizes the work of [132] for MSDA,

$$\min_{\alpha, g_\theta, h_\xi} \frac{1}{k}\sum_{k=1}^{K}\hat{\mathcal{R}}_{P_{S_k}}(h_\xi \circ g_\theta) + \mathcal{T}_{c_{\theta,\xi}}(\hat{P}_T^{h_\xi}, \sum_{k=1}^{K}\alpha_k \hat{P}_{S_k}).$$

Heterogeneous Domain Adaptation (HDA) is a DA problem in which source and target domains are incomparable, that is, $\mathbf{X}^{(P_S)} \in \mathbb{R}^{n_S \times d_S}$ and $\mathbf{X}^{(P_T)} \in \mathbb{R}^{n_T \times d_T}$, $n_S \neq n_T$ and $d_S \neq d_T$. To that end, [138] relies on the GW OT formalism (see sec. 3.5),

$$(\gamma^{(s)}, \gamma^{(f)}) = \min_{\substack{\gamma^{(s)} \in \Gamma(\mathbf{w}_S, \mathbf{w}_T) \\ \gamma^{(f)} \in \Gamma(\mathbf{v}_S, \mathbf{v}_T)}} \sum_{i,j,k,l} L(\mathbf{X}_{i,k}^{(P_S)}, \mathbf{X}_{j,l}^{(P_T)})\gamma_{ij}^{(s)}\gamma_{ij}^{(f)},$$

where $\gamma^{(s)} \in \mathbb{R}^{n_S \times n_T}$ and $\gamma^{(f)} \in \mathbb{R}^{d_S \times d_T}$ are OT plans between *samples* and *features*, respectively. Using $\gamma^{(s)}$, one can estimate labels on the target domain using label propagation [139], $\hat{\mathbf{Y}}^{(P_T)} = \text{diag}(\mathbf{p}_T)^{-1}(\gamma^{(s)})^T\mathbf{Y}^{(P_S)}$.

Transferability concerns the task of predicting whether TL will be successful. From a theoretical perspective, different IPMs bound the target risk $\mathcal{R}_{P_T}$ by the source risk $\mathcal{R}_{P_S}$, such as the $\mathcal{H}$, Wasserstein and MMD distances (see e.g., [130], [140], [141]). Furthermore there is a practical interest in accurately measuring transferability *a priori*, before training or fine-tuning a network. A candidate measure coming from OT is the Wasserstein distance, but in its original formulation it only takes features into account. [142] propose the Optimal Transport Dataset Distance (OTDD), which relies on a Wasserstein-distance based metric on the label space,

$$\text{OTDD}(P_S, P_T) = \underset{\gamma \in \Gamma(P_S,P_T)}{\arg\min} \underset{(\mathbf{z}_1,\mathbf{z}_2)\sim\gamma}{\mathbb{E}}[c(\mathbf{z}_1,\mathbf{z}_2)],$$
$$c(\mathbf{z}_1,\mathbf{z}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 + W_2(P_{S,y_1}, P_{T,y_2})^2,$$

where $\mathbf{z} = (\mathbf{x}, y)$, and $P_{S,y}$ is the conditional distribution $P_S(X|Y = y)$. As the authors show in [142], this distance is highly correlated with transferability and can even be used to interpolate between datasets with different classes [143].

7 Reinforcement Learning

Reinforcement Learning (RL) deals with dynamic learning scenarios and sequential decision-making. Following [144], one assumes an environment modeled through a Markov Decision Process (MDP), which is a 5-tuple $(\mathcal{S}, \mathcal{A}, P, R, \rho_0, \lambda)$ of a state space $\mathcal{S}$, an action space $\mathcal{A}$, a transition distribution $P : \mathcal{S}\times\mathcal{A}\times\mathcal{S} \rightarrow \mathbb{R}$, a reward function $R : \mathcal{S}\times\mathcal{A} \rightarrow \mathbb{R}$, a distribution over the initial state ρ_0, and a discount factor $\lambda \in (0, 1)$.

RL revolves around an agent acting on the state space, changing the state of the environment. The actions are chosen according to a policy $\pi : \mathcal{S} \rightarrow \mathbb{R}$, which is a distribution $\pi(\cdot|s)$ over actions $a \in \mathcal{A}$, for $s \in \mathcal{S}$. Policies can be evaluated according their average returns,

$$\eta(\pi) = \underset{P,\pi}{\mathbb{E}}\biggl[\sum_{t=0}^{\infty}\lambda^t R(s_t, a_t)\biggr]. \tag{45}$$

Furthermore, under π, one can evaluate states and state-action pairs through V_π and Q_π,

$$\begin{aligned} V_\pi(s) &= \underset{P,\pi}{\mathbb{E}}\biggl[\sum_{\ell=0}^{\infty}\lambda^\ell R(s_{t+\ell}, a_{t+\ell})|s_t = s\biggr], \\ Q_\pi(s,a) &= \underset{P,\pi}{\mathbb{E}}\biggl[\sum_{\ell=0}^{\infty}\lambda^\ell R(s_{t+\ell}, a_{t+\ell})|s_t = s, a_t = a\biggr], \end{aligned} \tag{46}$$

where the expectation over P and π corresponds to $s_0 \sim \rho_0$, $a_t \sim \pi(\cdot|s_t)$, and $s_{t+1} \sim P(\cdot|s_t, a_t)$. These quantities are related through Bellman's equation [145],

$$\begin{aligned} \mathcal{T}_\pi V_\pi(s_t) &= \underset{P,\pi}{\mathbb{E}}[R(s_t,a_t)] + \gamma \underset{P,\pi}{\mathbb{E}}[V_\pi(s)], \\ \mathcal{T}_\pi Q_\pi(s_t,a_t) &= \underset{P,\pi}{\mathbb{E}}[R(s_t,a_t)] + \gamma \underset{P,\pi}{\mathbb{E}}[Q_\pi(s,a)], \end{aligned} \tag{47}$$

where $\mathcal{T}_\pi$ is called Bellman operator. Q_π can be learned *from experience* (i.e., tuples (s, a, r, s')) through Q-Learning [146]. For a learning rate $\alpha_t > 0$, the updates are as follows,

$$Q_\pi(s,a) \leftarrow (1-\alpha_t)Q_\pi(s,a) + \alpha_t\biggl(r + \lambda V_\pi(s')\biggr). \tag{48}$$

7.1 Distributional Reinforcement Learning

Distributional Reinforcement Learning (DRL) [147], [148], [149] differs from traditional RL by considering uncertainty over returns. In this context, [150] proposed studying the random return Z_π, such that $Q_\pi(s_t, a_t) = \underset{P,\pi}{\mathbb{E}}[Z_\pi(s_t, a_t)]$, where the expectation is taken in the sense of equations 46. [150] defines DRL by analogy with equation 47,

$$Z_\pi(s_t,a_t) \overset{D}{=} R(s_t,a_t) + \lambda Z_\pi(s_{t+1},a_{t+1}),$$

where $\overset{D}{=}$ means equality in distribution. Note that Z_π is a distribution over returns, hence a distribution over $\mathbb{R}$. In DRL, one has mainly two choices for parametrizing Z_π, which rely on an empirical approximation,

$$Z_\theta(x) = \sum_{i=1}^{n} z_i \delta(x - x_i),$$

where one focuses either on z_i [150], or the x_i [151]. The optimal θ is found by minimizing a notion of discrepancy between Z_θ and $T_\pi Z_\theta$. OT contributes to DRL precisely at this point: one uses $W_p(Z_\theta, T_\pi Z_\theta)$.

Concerning the parametrization choice, let $\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^n$. [150] suggests to fix $\{x_i\}_{i=1}^n$ and estimate $\mathbf{z} = \text{softmax}(\theta(s, a))$. This poses technical challenges, as the stochastic gradients of W_p are biased. To tackle this issue, [151] proposes fixing $z_i := n^{-1}$ and estimating $x_i = (\theta(s, a))_i$. This choice presents 2 advantages; (i) Discretizing Z_θ's support is more flexible as the support is now free, potentially leading to more accurate predictions; (ii) It allows the use of the Wasserstein distance as loss without suffering from biased gradients.

7.2 Bayesian Reinforcement Learning

Similarly to DRL, Bayesian Reinforcement Learning (BRL) [152] adopts a distributional view reflecting the uncertainty over a given variable. In the remainder of this section, we discuss the Wasserstein Q-Learning (WQL) algorithm of [153], a modification of the Q-Learning algorithm of [146] (eq. 48). For a family of distributions $\mathcal{Q}$ (e.g., Gaussian distributions), this strategy considers a distribution $Q(s, a) \in \mathcal{Q}$, called $Q-$posterior, which represents the posterior distribution of the $Q-$function estimate. For each state there is a $V-$posterior $V(s)$, defined in terms of Wasserstein barycenters,

$$V(s) \in \underset{V\in\mathcal{Q}}{\arg\inf} \underset{a\sim\pi(\cdot|s)}{\mathbb{E}}[W_2(V, Q(s,a))^2]. \tag{49}$$

This formulation shows an alternative, continuous view of Wasserstein barycenters, as one has infinitely many distributions $Q(\cdot, a)$ over states, for $a \sim \pi(\cdot|s)$.

Upon a transition (s, a, s', r), [153] defines the Wasserstein Temporal Difference, which is the distributional analogous to equation 48,

$$Q_{t+1}(s,a) \in \mathcal{B}([1-\alpha_t, \alpha_t]; \{Q_t(s,a), \mathcal{T}_\pi Q_t(s,a)\}) \tag{50}$$

where $\alpha_t > 0$ is the learning rate, whereas $\mathcal{T}_\pi Q_t = r + \lambda V_t$. For specific families $\mathcal{Q}$ (e.g., Gaussian distributions), eqs. 49 and 50 have an analytical solution (see [153, Table 1]).

7.3 Policy Optimization

Policy Optimization (PO) focuses on searching for a policy π that maximizes $\eta(\pi)$. In this section we focus on gradient-based approaches (e.g., [154]), who parametrize $\pi = \pi_\theta$ so that training consists on maximizing $\eta(\theta) = \eta(\pi_\theta)$. However, as [155] remarks, these algorithms suffer from high variance and slow convergence. [155] thus propose a distributional view. Let P be a distribution over θ, they propose the following optimization problem,

$$P^\star = \underset{P}{\arg\max} \underset{\theta \sim P}{\mathbb{E}}[\eta(\pi_\theta)] - \alpha \text{KL}(P \| P_0), \tag{51}$$

where P_0 is a prior, and $\alpha > 0$ controls regularization. The minimum of equation 51 implies $P(\theta) \propto \exp(\eta(\pi_\theta)/\alpha)$. From a Bayesian perspective, $P(\theta)$ is the posterior distribution, whereas $\exp(\eta(\pi_\theta)/\alpha)$ is the likelihood function. As discussed in [156], this formulation can be written in terms of gradient flows in the space of probability distributions (see [157], [158] for further details). First, consider the functional,

$$F(P) = -\int P(\theta) \log P_0(\theta) d\theta + \int P(\theta) \log P(\theta) d\theta.$$

For Itó-diffusion [159], optimizing F w.r.t. P becomes,

$$P_{k+1}^{(h)} = \underset{P}{\arg\min}\, \text{KL}(P \| P_0) + \frac{W_2^2(P, P_k^{(h)})}{2h}, \tag{52}$$

which corresponds to the Jordan-Kinderlehrer-Otto (JKO) scheme [159]. In this context, [156] introduces 2 formulations for learning an optimal policy: indirect and direct learning. In the first case, one defines the gradient flow for equation 52, in terms of θ. This setting is particularly suited when the parameter space is low dimensional. In the second case, one uses policies directly into the described framework, which is related to [160].

8 Final Remarks

8.1 Challenges

Curse of Dimensionality. A known theoretical fact is that OT estimation becomes harder in high dimensions [161], [162], i.e., $|W_p(P,Q) - W_p(\hat{P},\hat{Q})| \leq \mathcal{O}(n^{-1/d})$. We illustrate this issue by extending the experiments of [28] in Figure 15. Especially, we compare W_2, $W_{2,\epsilon}$, $S_{2,\epsilon}$ and SW_2 for $P = Q = \mathcal{U}([0,1]^d)$, i.e., the uniform distribution over the $d-$dimensional hypercube. Naturally, $W_p(P,Q) = 0$. We then sample $\mathbf{X}^{(P)} \sim P$ and $\mathbf{X}^{(Q)} \sim Q$ independently. We analyze these values as a function of the number of samples n, and the dimension of the data d. Note that, except for SW_2, the rate of estimation decays as we increase d. This challenge is key in several areas of ML, as data commonly lives in a high-dimensional space.

Computational Complexity. When computing exact OT, one has a $\mathcal{O}(n^3 \log n)$ complexity over the amount of samples coming from the expensive linear program. This issue has been partially alleviated with the Sinkhorn algorithm of [25], which have $\mathcal{O}(n^2)$ complexity *per iteration*, and is amenable on GPU. Another direction consists of breaking probability distributions into L 1$-$D slices, which reduces the OT problem to L sorting problems, which has $\mathcal{O}(n \log n)$ complexity. Nonetheless, on one hand, for a large number

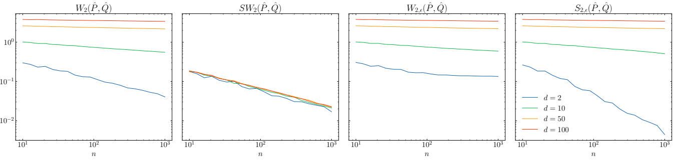

Fig. 15: Estimation of Wasserstein distances with finite samples as a function of number of samples n, and dimensions d. Overall, the plug-in empirical estimator $W_2(\hat{P},\hat{Q})$ suffers from the curse of dimensionality, as estimation becomes harder in high dimensions. This issue can be alleviated through alternative estimators, such as sliced Wasserstein or entropic OT.

of Sinkhorn iterations, exact OT may be more efficient. The same remark can be made concerning the number of projections for the SW distance. On the other hand, the Sinkhorn divergence is smoother than the exact Wasserstein distance [92], which is desirable for optimization.

In table 1 we show a summary of *time* and *sample* complexities for OT estimators. While the first concerns computational or time of execution, the second concerns the number of samples needed to accurately estimate the Wasserstein distance W_p.

TABLE 1: Time and sample complexity of empirical OT estimators in terms of the number of samples n. † indicates complexity of a single iteration; * indicates that the complexity is affected by samples dimension.

Reference	Estimator	Time Complexity	Sample Complexity
[161]	$W_p(\hat{P},\hat{Q})$	$\mathcal{O}(n^3 \log n)$	$\mathcal{O}(n^{-1/d})$
[28]	$S_{p,\epsilon}(\hat{P},\hat{Q})$	$\mathcal{O}(n^2)^\dagger$	$\mathcal{O}(n^{-1/2}(1+\epsilon^{\lfloor d/2 \rfloor}))$
[163]	$\text{SRW}_k(\hat{P},\hat{Q})$	$\mathcal{O}(n^2)^{\dagger,*}$	$\mathcal{O}(n^{-1/k})$
[98]	$\text{SW}_p(\hat{P},\hat{Q})$	$\mathcal{O}(n \log n)^*$	$\mathcal{O}(n^{-1/2})$
[42]	$FW_{K,2}(\hat{P},\hat{Q})$	$\mathcal{O}(n^2)^\dagger$	$\mathcal{O}(n^{-1/2})$
[53]	$\mathcal{L}_{\text{MBOT}}^{(k,m)}(\hat{P},\hat{Q})$	$\mathcal{O}(km^3 \log m)$	$\mathcal{O}(n^{-1/2})$

8.2 Future Trends

ML for OT. Several works [44], [45], [46], [47] consider introducing ML methodology in OT theory, primarily through NNs. The main advantage of doing so is scaling OT methods to larger datasets and higher dimensions. However, without a careful analysis, the resulting methods may end up not estimating OT (e.g., [44], [164]), even if the method successfully performs the ML task, as in the case of generative modeling [91], which suggests an apparent separation between OT estimation and ML tasks. However, recent approaches [46], [47] pose OT estimation as a learning task. This trend shows an interplay between OT and ML practice.

Sliced Wasserstein distances. The SW distance has been generalized by a number of works [33], [165], [166]. For instance, while [33] proposes a non-linear slicing method through NNs, [165] slices distributions through convolutions. Furthermore, [166] define the SW distance over the sphere $\mathbb{S}^{d-1}$. Further research can focus on the idea of SW distances over manifolds.

Decentralized and Private OT. In recent ML practice, researchers considered distributed learning over several devices or clients without directly communicating data [167], known as *federated learning*. In this context, *differential privacy* a strong guarantee for protecting clients' data [168]. Recent advances in OT devise ways to privately [169], [170] and federated [171] computing the Wasserstein distance, as well as distributed MSDA strategies [172]. These works serve as a starting point for OT-inspired federated strategies in other fields of ML.

Supervised Learning. As discussed in section 4, OT contributes in two ways: defining a loss that considers semantic similarities between classes and defining robust alternatives to ERM. Future works could consider the construction of label embeddings (vector representations) for classes so that these embeddings capture the geometry of their corresponding distributions, i.e., $P(X|Y = y)$.

Generative Modeling. This subject has been one of the most active OTML topics. This phenomenon is evidenced by the impact of papers such as [44] and [87] had on how generative models are understood and trained [85]. Future works can focus on using extensions to OT as a loss. For instance, one can devise outlier-robust generative models through unbalanced or partial OT. Furthermore, one can devise generative graph models with the FGW distance.

Dictionary Learning. OT provides a rich theory for dictionary learning when objects have an underlying probabilistic interpretation, such as histograms. In this sense, one can perform dictionary learning in Wasserstein space by using the Wasserstein distance as a loss function and using Wasserstein barycenters for combining atoms. This idea is recurrent in ML, e.g., images and text [113], graphs [114], [115] and domain adaptation [173]. Future contributions can focus on the manifold of interpolations of atoms in the Wasserstein space and the interpretability of coefficients $\mathbf{a}_i$.

Clustering. As shown in section 5.3, OT provides a probabilistic framework for clustering algorithms (e.g., k-Means [125]). Furthermore, OT contributed to two clustering settings: co-clustering and GMMs. In the first case, [127] relies on the GW formalism for finding clusters over samples and features. In the second case, [96] uses the SW distance for learning GMMs. This metric works well in high dimensions, is easy to compute, and is robust to initialization. Future works can consider the theoretical analysis of the approach of [96], such as proving convergence and robustness to initialization.

Transfer Learning. From a practical perspective, OT-based UDA has shown state-of-the-art performance in various fields, such as image classification [39], sentiment analysis [132], fault diagnosis [174], [175], and audio classification [136], [137]. Advances in the field include methods and architectures that can handle large-scale datasets, such as WDGRL [135] and DeepJDOT [51]. From a theoretical perspective, various works [130], [141] show that OT is an essential framework for *formalizing UDA*. This effectively consolidated OT as a valuable toolbox for transferring knowledge between different domains.

One may draw parallels between UDA, fairness, and generative modeling in a broader context. In the first case, the random repair strategy of [80] is conceptually similar to OTDA [39]. This similarity suggests that theoretical results on the optimal amount of repaired data can be derived, similarly to [141, Theorem 3]. Furthermore, parallel between (DANN, WDGRL) and (GAN, WGAN), which shows an interesting interplay between the two fields.

Reinforcement Learning. Reinforcement Learning. The distributional RL framework of [150] re-introduced previous ideas in modern RL, such as formulating the goal of an agent in terms of distributions over rewards. This idea turned out to be successful, as the authors' method surpassed the state-of-the-art in a variety of game benchmarks. Furthermore, [176] studies how this framework relates to dopamine-based RL, highlighting this formulation's importance.

In a similar spirit, [153] proposes a distributional method for propagating uncertainty of the value function. Here, it is important to highlight that while [150] focuses on the randomness of the reward function, [153] studies the stochasticity of value functions. In this sense, the authors propose a variant of Q-Learning, called WQL, that leverages Wasserstein barycenters for updating the $Q-$function distribution. From a practical perspective, the WQL algorithm is theoretically grounded and has good performance. Further research can focus on the common points between [150], [151], and [153], either by establishing a common framework or by comparing the 2 methodologies empirically. Finally, [156] explores policy optimization as gradient flows in the space of distributions. This formalism can be understood as the continuous-time counterpart of gradient descent.

8.3 Conclusion

Optimal transport serves as a formal toolkit for probabilistic machine learning. In this survey, we cover applications to supervised, unsupervised, transfer, and reinforcement learning settings, as well as *how to compute it*. In a nutshell, it is used for computing losses or manipulating probability distributions. On the one hand, the main attractive points of this theory are its flexibility and useful theoretical properties. On the other hand, two challenges remain relevant in the context of machine learning: OT is difficult to estimate in high dimensions, and it is computationally heavy. Finally, OT has positively impacted the probabilistic machine learning landscape. Conversely, solutions coming from machine learning begin to influence OT, such as using neural networks to approximate OT maps and plans.

References

[1] G. Monge, "Mémoire sur la théorie des déblais et des remblais," *Histoire de l'Académie Royale des Sciences de Paris*, 1781.

[2] L. Kantorovich, "On the transfer of masses (in russian)," in *Doklady Akademii Nauk*, vol. 37, no. 2, 1942, pp. 227–229.

[3] C. Villani, *Optimal transport: old and new*. Springer, 2009, vol. 338.

[4] U. Frisch, S. Matarrese, R. Mohayaee, and A. Sobolevski, "A reconstruction of the initial conditions of the universe by optimal mass transportation," *Nature*, vol. 417, no. 6886, pp. 260–262, 2002.

[5] Y. Rubner, C. Tomasi, and L. J. Guibas, "The earth mover's distance as a metric for image retrieval," *International journal of computer vision*, vol. 40, no. 2, pp. 99–121, 2000.

[6] L. Ambrosio and N. Gigli, "A user's guide to optimal transport," in *Modelling and optimisation of flows on networks*. Springer, 2013, pp. 1–155.

[7] S. Kolouri, S. R. Park, M. Thorpe, D. Slepcev, and G. K. Rohde, "Optimal mass transport: Signal processing and machine-learning applications," *IEEE signal processing magazine*, vol. 34, no. 4, pp. 43–59, 2017.

[8] J. Solomon, "Optimal transport on discrete domains," *AMS Short Course on Discrete Differential Geometry*, 2018.

[9] B. Lévy and E. L. Schwindt, "Notions of optimal transport theory and how to implement them on a computer," *Computers & Graphics*, vol. 72, pp. 135–148, 2018.

[10] G. Peyré, M. Cuturi *et al.*, "Computational optimal transport: With applications to data science," *Foundations and Trends® in Machine Learning*, vol. 11, no. 5-6, pp. 355–607, 2019.

[11] R. Flamary, "Transport optimal pour l'apprentissage statistique," *Habilitation à diriger des recherches. Université Côte d'Azur*, 2019.

[12] F. Santambrogio, "Optimal transport for applied mathematicians," *Birkäuser, NY*, vol. 55, no. 58-63, p. 94, 2015.

[13] G. Monge, *Mémoire sur le calcul intégral des équations aux différences partielles*. Imprimerie royale, 1784.

[14] Y. Brenier, "Polar factorization and monotone rearrangement of vector-valued functions," *Communications on pure and applied mathematics*, vol. 44, no. 4, pp. 375–417, 1991.

[15] C. Villani, *Topics in optimal transportation*. American Mathematical Soc., 2021, vol. 58.

[16] R. J. McCann, "A convexity principle for interacting gases," *Advances in mathematics*, vol. 128, no. 1, pp. 153–179, 1997.

[17] M. Agueh and G. Carlier, "Barycenters in the wasserstein space," *SIAM Journal on Mathematical Analysis*, vol. 43, no. 2, pp. 904–924, 2011.

[18] A. Müller, "Integral probability metrics and their generating classes of functions," *Advances in Applied Probability*, vol. 29, no. 2, pp. 429–443, 1997.

[19] I. Csiszár, "Information-type measures of difference of probability distributions and indirect observation," *studia scientiarum Mathematicarum Hungarica*, vol. 2, pp. 229–318, 1967.

[20] B. K. Sriperumbudur, K. Fukumizu, A. Gretton, B. Schölkopf, and G. R. Lanckriet, "On the empirical estimation of integral probability metrics," *Electronic Journal of Statistics*, vol. 6, pp. 1550–1599, 2012.

[21] A. Gretton, K. M. Borgwardt, M. Rasch, B. Schölkopf, and A. J. Smola, "A kernel approach to comparing distributions," in *Proceedings of the National Conference on Artificial Intelligence*, vol. 22, no. 2. Menlo Park, CA; Cambridge, MA; London; AAAI Press; MIT Press; 1999, 2007, p. 1637.

[22] R. Flamary, N. Courty, A. Gramfort, M. Z. Alaya, A. Boisbunon, S. Chambon, L. Chapel, A. Corenflos, K. Fatras, N. Fournier *et al.*, "Pot: Python optimal transport," *Journal of Machine Learning Research*, vol. 22, no. 78, pp. 1–8, 2021.

[23] M. Cuturi, L. Meng-Papaxanthos, Y. Tian, C. Bunne, G. Davis, and O. Teboul, "Optimal transport tools (ott): A jax toolbox for all things wasserstein," *arXiv preprint arXiv:2201.12324*, 2022.

[24] G. B. Dantzig, "Reminiscences about the origins of linear programming," in *Mathematical programming the state of the art*. Springer, 1983, pp. 78–86.

[25] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," *Advances in neural information processing systems*, vol. 26, pp. 2292–2300, 2013.

[26] A. Genevay, G. Peyré, and M. Cuturi, "Learning generative models with sinkhorn divergences," in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2018, pp. 1608–1617.

[27] G. Luise, A. Rudi, M. Pontil, and C. Ciliberto, "Differential properties of sinkhorn approximation for learning with wasserstein distance," *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[28] A. Genevay, L. Chizat, F. Bach, M. Cuturi, and G. Peyré, "Sample complexity of sinkhorn divergences," in *The 22nd international conference on artificial intelligence and statistics*. PMLR, 2019, pp. 1574–1583.

[29] J. Rabin, G. Peyré, J. Delon, and M. Bernot, "Wasserstein barycenter and its application to texture mixing," in *International Conference on Scale Space and Variational Methods in Computer Vision*. Springer, 2011, pp. 435–446.

[30] N. Bonneel, J. Rabin, G. Peyré, and H. Pfister, "Sliced and radon wasserstein barycenters of measures," *Journal of Mathematical Imaging and Vision*, vol. 51, no. 1, pp. 22–45, 2015.

[31] S. Kolouri, S. R. Park, and G. K. Rohde, "The radon cumulative distribution transform and its application to image classification," *IEEE transactions on image processing*, vol. 25, no. 2, pp. 920–934, 2015.

[32] I. Deshpande, Y.-T. Hu, R. Sun, A. Pyrros, N. Siddiqui, S. Koyejo, Z. Zhao, D. Forsyth, and A. G. Schwing, "Max-sliced wasserstein distance and its use for gans," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 10 648–10 656.

[33] S. Kolouri, K. Nadjahi, U. Simsekli, R. Badeau, and G. Rohde, "Generalized sliced wasserstein distances," *Advances in neural information processing systems*, vol. 32, 2019.

[34] T. T. Nguyen, S. Gupta, and S. Venkatesh, "Distributional reinforcement learning via moment matching," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2021.

[35] F.-P. Paty and M. Cuturi, "Subspace robust wasserstein distances," in *International conference on machine learning*. PMLR, 2019, pp. 5072–5081.

[36] T. Lin, C. Fan, N. Ho, M. Cuturi, and M. Jordan, "Projection robust wasserstein distance and riemannian optimization," *Advances in neural information processing systems*, vol. 33, pp. 9383–9397, 2020.

[37] M. Huang, S. Ma, and L. Lai, "A riemannian block coordinate descent method for computing the projection robust wasserstein distance," in *International Conference on Machine Learning*. PMLR, 2021, pp. 4446–4455.

[38] S. Ferradans, N. Papadakis, G. Peyré, and J.-F. Aujol, "Regularized discrete optimal transport," *SIAM Journal on Imaging Sciences*, vol. 7, no. 3, pp. 1853–1882, 2014.

[39] N. Courty, R. Flamary, D. Tuia, and A. Rakotomamonjy, "Optimal transport for domain adaptation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 9, pp. 1853–1865, 2017.

[40] D. Alvarez-Melis, T. Jaakkola, and S. Jegelka, "Structured optimal transport," in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2018, pp. 1771–1780.

[41] L. Lovász, "Mathematical programming–the state of the art," 1983.

[42] A. Forrow, J.-C. Hütter, M. Nitzan, P. Rigollet, G. Schiebinger, and J. Weed, "Statistical optimal transport via factored couplings," in *The 22nd International Conference on Artificial Intelligence and Statistics*. PMLR, 2019, pp. 2454–2465.

[43] J. E. Cohen and U. G. Rothblum, "Nonnegative ranks, decompositions, and factorizations of nonnegative matrices," *Linear Algebra and its Applications*, vol. 190, pp. 149–168, 1993.

[44] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *International conference on machine learning*. PMLR, 2017, pp. 214–223.

[45] V. Seguy, B. B. Damodaran, R. Flamary, N. Courty, A. Rolet, and M. Blondel, "Large scale optimal transport and mapping estimation," in *International Conference on Learning Representations*, 2018.

[46] A. Makkuva, A. Taghvaei, S. Oh, and J. Lee, "Optimal transport mapping via input convex neural networks," in *International Conference on Machine Learning*. PMLR, 2020, pp. 6672–6681.

[47] A. Korotin, D. Selikhanovych, and E. Burnaev, "Neural optimal transport," in *The Eleventh International Conference on Learning Representations*, 2023.

[48] B. Amos, L. Xu, and J. Z. Kolter, "Input convex neural networks," in *International Conference on Machine Learning*. PMLR, 2017, pp. 146–155.

[49] G. Montavon, K.-R. Müller, and M. Cuturi, "Wasserstein training of restricted boltzmann machines." in *NIPS*, 2016, pp. 3711–3719.

[50] M. Sommerfeld, J. Schrieber, Y. Zemel, and A. Munk, "Optimal transport: Fast probabilistic approximation with exact solvers." *J. Mach. Learn. Res.*, vol. 20, pp. 105–1, 2019.

[51] B. B. Damodaran, B. Kellenberger, R. Flamary, D. Tuia, and N. Courty, "Deepjdot: Deep joint distribution optimal transport for unsupervised domain adaptation," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 447–463.

[52] E. Bernton, P. E. Jacob, M. Gerber, and C. P. Robert, "On parameter estimation with the wasserstein distance," *Information and Inference: A Journal of the IMA*, vol. 8, no. 4, pp. 657–676, 2019.

[53] K. Fatras, Y. Zine, R. Flamary, R. Gribonval, and N. Courty, "Learning with minibatch wasserstein: asymptotic and gradient properties," in *AISTATS*, 2020.

[54] K. Fatras, T. Sejourne, R. Flamary, and N. Courty, "Unbalanced minibatch optimal transport; applications to domain adaptation," in *International Conference on Machine Learning*. PMLR, 2021, pp. 3186–3197.

[55] M. Liero, A. Mielke, and G. Savaré, "Optimal entropy-transport problems and a new hellinger–kantorovich distance between positive measures," *Inventiones mathematicae*, vol. 211, no. 3, pp. 969–1117, 2018.

[56] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced optimal transport problems," *Mathematics of Computation*, vol. 87, no. 314, pp. 2563–2609, 2018.

[57] L. Chapel, M. Z. Alaya, and G. Gasso, "Partial optimal tranport with applications on positive-unlabeled learning," *Advances in Neural Information Processing Systems*, vol. 33, pp. 2903–2913, 2020.

[58] K. Nguyen, D. Nguyen, T. Pham, N. Ho *et al.*, "Improving minibatch optimal transport via partial transportation," in *International Conference on Machine Learning*. PMLR, 2022, pp. 16 656–16 690.

[59] F. Mémoli, "Gromov–wasserstein distances and the metric approach to object matching," *Foundations of computational mathematics*, vol. 11, no. 4, pp. 417–487, 2011.

[60] V. Titouan, N. Courty, R. Tavenard, and R. Flamary, "Optimal transport for structured data with application on graphs," in *International Conference on Machine Learning*. PMLR, 2019, pp. 6275–6284.

[61] E. M. Loiola, N. M. M. De Abreu, P. O. Boaventura-Netto, P. Hahn, and T. Querido, "A survey for the quadratic assignment problem," *European journal of operational research*, vol. 176, no. 2, pp. 657–690, 2007.

[62] G. Peyré, M. Cuturi, and J. Solomon, "Gromov-wasserstein averaging of kernel and distance matrices," in *International Conference on Machine Learning*. PMLR, 2016, pp. 2664–2672.

[63] J. Solomon, G. Peyré, V. G. Kim, and S. Sra, "Entropic metric alignment for correspondence problems," *ACM Transactions on Graphics (ToG)*, vol. 35, no. 4, pp. 1–13, 2016.

[64] N. Gozlan, C. Roberto, P.-M. Samson, and P. Tetali, "Kantorovich duality for general transport costs and applications," *Journal of Functional Analysis*, vol. 273, no. 11, pp. 3327–3405, 2017.

[65] R. T. Rockafellar, "Integral functionals, normal integrands and measurable selections," in *Nonlinear Operators and the Calculus of Variations: Summer School Held in Bruxelles 8–19 September 1975*. Springer, 2006, pp. 157–207.

[66] V. Vapnik, *The nature of statistical learning theory*. Springer science & business media, 2013.

[67] C. M. Bishop and N. M. Nasrabadi, *Pattern recognition and machine learning*. Springer, 2006, vol. 4, no. 4.

[68] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.

[69] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," in *Advances in neural information processing systems*, 2013, pp. 3111–3119.

[70] C. Frogner, C. Zhang, H. Mobahi, M. Araya, and T. A. Poggio, "Learning with a wasserstein loss," *Advances in Neural Information Processing Systems*, vol. 28, pp. 2053–2061, 2015.

[71] X. Liu, Y. Han, S. Bai, Y. Ge, T. Wang, X. Han, S. Li, J. You, and J. Lu, "Importance-aware semantic segmentation in self-driving with discrete wasserstein training," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 07, 2020, pp. 11 629–11 636.

[72] X. Liu, W. Ji, J. You, G. E. Fakhri, and J. Woo, "Severity-aware semantic segmentation with reinforced wasserstein training," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 12 566–12 575.

[73] K. Fatras, B. B. Damodaran, S. Lobry, R. Flamary, D. Tuia, and N. Courty, "Wasserstein adversarial regularization for learning with label noise," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, pp. 7296–7306, 2022.

[74] H. Rahimian and S. Mehrotra, "Distributionally robust optimization: A review," *arXiv preprint arXiv:1908.05659*, 2019.

[75] D. Kuhn, P. M. Esfahani, V. A. Nguyen, and S. Shafieezadeh-Abadeh, "Wasserstein distributionally robust optimization: Theory and applications in machine learning," in *Operations research & management science in the age of analytics*. Informs, 2019, pp. 130–166.

[76] S. Shafieezadeh-Abadeh, D. Kuhn, and P. M. Esfahani, "Regularization via mass transportation," *Journal of Machine Learning Research*, vol. 20, no. 103, pp. 1–68, 2019.

[77] M. B. Zafar, I. Valera, M. Gomez Rodriguez, and K. P. Gummadi, "Fairness beyond disparate treatment & disparate impact: Learning classification without disparate mistreatment," in *Proceedings of the 26th international conference on world wide web*, 2017, pp. 1171–1180.

[78] S. Caton and C. Haas, "Fairness in machine learning: A survey," *ACM Computing Surveys*, 2020.

[79] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," in *Proceedings of the 3rd innovations in theoretical computer science conference*, 2012, pp. 214–226.

[80] P. Gordaliza, E. Del Barrio, G. Fabrice, and J.-M. Loubes, "Obtaining fairness using optimal transport theory," in *International Conference on Machine Learning*. PMLR, 2019, pp. 2357–2365.

[81] L. Oneto, M. Donini, G. Luise, C. Ciliberto, A. Maurer, and M. Pontil, "Exploiting mmd and sinkhorn divergences for fair and transferable representation learning." in *NeurIPS*, 2020.

[82] S. Chiappa, R. Jiang, T. Stepleton, A. Pacchiano, H. Jiang, and J. Aslanides, "A general approach to fairness with optimal transport." in *AAAI*, 2020, pp. 3633–3640.

[83] A. Genevay, G. Peyré, and M. Cuturi, "Gan and vae from an optimal transport point of view," *arXiv preprint arXiv:1706.01807*, 2017.

[84] N. Lei, K. Su, L. Cui, S.-T. Yau, and X. D. Gu, "A geometric view of optimal transportation and generative model," *Computer Aided Geometric Design*, vol. 68, pp. 1–21, 2019.

[85] F. Bassetti, A. Bodini, and E. Regazzini, "On minimum kantorovich distance estimators," *Statistics & probability letters*, vol. 76, no. 12, pp. 1298–1302, 2006.

[86] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in neural information processing systems*, vol. 27, 2014.

[87] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of wasserstein gans," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017.

[88] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *International Conference on Learning Representations*, 2018.

[89] T. Pinetz, D. Soukup, and T. Pock, "On the estimation of the wasserstein distance in generative models," in *German Conference on Pattern Recognition*. Springer, 2019, pp. 156–170.

[90] A. Mallasto, G. Montúfar, and A. Gerolin, "How well do wgans estimate the wasserstein metric?" *arXiv preprint arXiv:1910.03875*, 2019.

[91] J. Stanczuk, C. Etmann, L. M. Kreusser, and C.-B. Schönlieb, "Wasserstein gans work because they fail (to approximate the wasserstein distance)," *arXiv preprint arXiv:2103.01678*, 2021.

[92] J. Feydy, T. Séjourné, F.-X. Vialard, S.-i. Amari, A. Trouvé, and G. Peyré, "Interpolating between optimal transport and mmd using sinkhorn divergences," in *The 22nd International Conference on Artificial Intelligence and Statistics*. PMLR, 2019, pp. 2681–2690.

[93] Y. Xie, X. Wang, R. Wang, and H. Zha, "A fast proximal point method for computing exact wasserstein distance," in *Uncertainty in artificial intelligence*. PMLR, 2020, pp. 433–453.

[94] D. Bertsekas, *Network optimization: continuous and discrete models*. Athena Scientific, 1998, vol. 8.

[95] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.

[96] S. Kolouri, G. K. Rohde, and H. Hoffmann, "Sliced wasserstein distance for learning gaussian mixture models," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 3427–3436.

[97] J. Wu, Z. Huang, D. Acharya, W. Li, J. Thoma, D. P. Paudel, and L. V. Gool, "Sliced wasserstein generative models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 3713–3722.

[98] K. Nadjahi, A. Durmus, U. Simsekli, and R. Badeau, "Asymptotic guarantees for learning generative models with the sliced-wasserstein distance," *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[99] K. Nadjahi, A. Durmus, L. Chizat, S. Kolouri, S. Shahrampour, and U. Simsekli, "Statistical and topological properties of sliced probability divergences," *Advances in Neural Information Processing Systems*, vol. 33, pp. 20 802–20 812, 2020.

[100] D. P. Kingma and M. Welling, "Autoencoding variational bayes," in *International Conference on Learning Representations*, 2020.

[101] D. M. Blei, A. Kucukelbir, and J. D. McAuliffe, "Variational inference: A review for statisticians," *Journal of the American statistical Association*, vol. 112, no. 518, pp. 859–877, 2017.

[102] A. Makhzani, J. Shlens, N. Jaitly, I. Goodfellow, and B. Frey, "Adversarial autoencoders," *arXiv preprint arXiv:1511.05644*, 2015.

[103] I. Tolstikhin, O. Bousquet, S. Gelly, and B. Schoelkopf, "Wasserstein auto-encoders," in *International Conference on Learning Representations*, 2018.

[104] G. Patrini, R. van den Berg, P. Forre, M. Carioni, S. Bhargav, M. Welling, T. Genewein, and F. Nielsen, "Sinkhorn autoencoders," in *Uncertainty in Artificial Intelligence*. PMLR, 2020, pp. 733–743.

[105] I. Kobyzev, S. Prince, and M. Brubaker, "Normalizing flows: An introduction and review of current methods," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020.

[106] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[107] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[108] C. Finlay, J.-H. Jacobsen, L. Nurbekyan, and A. Oberman, "How to train your neural ode: the world of jacobian and kinetic regularization," in *International Conference on Machine Learning*. PMLR, 2020, pp. 3154–3164.

[109] B. A. Olshausen and D. J. Field, "Sparse coding with an overcomplete basis set: A strategy employed by v1?" *Vision research*, vol. 37, no. 23, pp. 3311–3325, 1997.

[110] P. Paatero and U. Tapper, "Positive matrix factorization: A non-negative factor model with optimal utilization of error estimates of data values," *Environmetrics*, vol. 5, no. 2, pp. 111–126, 1994.

[111] R. Sandler and M. Lindenbaum, "Nonnegative matrix factorization with earth mover's distance metric," in *2009 IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 2009, pp. 1873–1880.

[112] A. Rolet, M. Cuturi, and G. Peyré, "Fast dictionary learning with a smoothed wasserstein loss," in *Artificial Intelligence and Statistics*. PMLR, 2016, pp. 630–638.

[113] M. A. Schmitz, M. Heitz, N. Bonneel, F. Ngole, D. Coeurjolly, M. Cuturi, G. Peyré, and J.-L. Starck, "Wasserstein dictionary learning: Optimal transport-based unsupervised nonlinear dictionary learning," *SIAM Journal on Imaging Sciences*, vol. 11, no. 1, pp. 643–678, 2018.

[114] C. Vincent-Cuaz, T. Vayer, R. Flamary, M. Corneli, and N. Courty, "Online graph dictionary learning," in *International Conference on Machine Learning*. PMLR, 2021, pp. 10 564–10 574.

[115] H. Xu, "Gromov-wasserstein factorization models for graph clustering," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 04, 2020, pp. 6478–6485.

[116] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent dirichlet allocation," *Journal of machine Learning research*, vol. 3, no. Jan, pp. 993–1022, 2003.

[117] M. Kusner, Y. Sun, N. Kolkin, and K. Weinberger, "From word embeddings to document distances," in *International conference on machine learning*. PMLR, 2015, pp. 957–966.

[118] V. Huynh, H. Zhao, and D. Phung, "Otlda: A geometry-aware optimal transport approach for topic modeling," *Advances in Neural Information Processing Systems*, vol. 33, pp. 18 573–18 582, 2020.

[119] M. Yurochkin, S. Claici, E. Chien, F. Mirzazadeh, and J. M. Solomon, "Hierarchical optimal transport for document representation," *Advances in neural information processing systems*, vol. 32, 2019.

[120] C. M. Bishop, *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Berlin, Heidelberg: Springer-Verlag, 2006.

[121] D. Pollard, "Quantization and the method of k-means," *IEEE Transactions on Information theory*, vol. 28, no. 2, pp. 199–205, 1982.

[122] G. Canas and L. Rosasco, "Learning probability measures with respect to optimal transport metrics," *Advances in Neural Information Processing Systems*, vol. 25, 2012.

[123] P. M. Gruber, "Optimum quantization and its applications," *Advances in Mathematics*, vol. 186, no. 2, pp. 456–497, 2004.

[124] S. Graf and H. Luschgy, *Foundations of quantization for probability distributions*. Springer, 2007.

[125] M. Cuturi and A. Doucet, "Fast computation of wasserstein barycenters," in *International conference on machine learning*. PMLR, 2014, pp. 685–693.

[126] E. Del Barrio, J. A. Cuesta-Albertos, C. Matrán, and A. Mayo-Íscar, "Robust clustering tools based on optimal transportation," *Statistics and Computing*, vol. 29, no. 1, pp. 139–160, 2019.

[127] C. Laclau, I. Redko, B. Matei, Y. Bennani, and V. Brault, "Co-clustering through optimal transport," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, D. Precup and Y. W. Teh, Eds., vol. 70. PMLR, 06–11 Aug 2017, pp. 1955–1964.

[128] B. Matei and S. Meignen, "Nonlinear cell-average multiscale signal representations: Application to signal denoising," *Signal processing*, vol. 92, no. 11, pp. 2738–2746, 2012.

[129] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on knowledge and data engineering*, vol. 22, no. 10, pp. 1345–1359, 2009.

[130] I. Redko, E. Morvant, A. Habrard, M. Sebban, and Y. Bennani, *Advances in domain adaptation theory*. Elsevier, 2019.

[131] M. Perrot, N. Courty, R. Flamary, and A. Habrard, "Mapping estimation for discrete optimal transport," *Advances in Neural Information Processing Systems*, vol. 29, pp. 4197–4205, 2016.

[132] N. Courty, R. Flamary, A. Habrard, and A. Rakotomamonjy, "Joint distribution optimal transportation for domain adaptation," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017.

[133] M. Ghifary, W. B. Kleijn, and M. Zhang, "Domain adaptive neural networks for object recognition," in *Pacific Rim international conference on artificial intelligence*. Springer, 2014, pp. 898–904.

[134] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *The journal of machine learning research*, vol. 17, no. 1, pp. 2096–2030, 2016.

[135] J. Shen, Y. Qu, W. Zhang, and Y. Yu, "Wasserstein distance guided representation learning for domain adaptation," in *Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.

[136] E. F. Montesuma and F. M. N. Mboula, "Wasserstein barycenter transport for acoustic adaptation," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 3405–3409.

[137] R. Turrisi, R. Flamary, A. Rakotomamonjy, and M. Pontil, "Multi-source domain adaptation via weighted joint distributions optimal transport," in *Uncertainty in Artificial Intelligence*. PMLR, 2022, pp. 1970–1980.

[138] I. Redko, T. Vayer, R. Flamary, and N. Courty, "Co-optimal transport," *Advances in Neural Information Processing Systems*, vol. 33, no. 17559-17570, 2020.

[139] I. Redko, N. Courty, R. Flamary, and D. Tuia, "Optimal transport for multi-source domain adaptation under target shift," in *The 22nd International Conference on Artificial Intelligence and Statistics*. PMLR, 2019, pp. 849–858.

[140] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, "A theory of learning from different domains," *Machine learning*, vol. 79, no. 1, pp. 151–175, 2010.

[141] I. Redko, A. Habrard, and M. Sebban, "Theoretical analysis of domain adaptation with optimal transport," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2017, pp. 737–753.

[142] D. Alvarez Melis and N. Fusi, "Geometric dataset distances via optimal transport," *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[143] D. Alvarez-Melis and N. Fusi, "Dataset dynamics via gradient flows in probability space," in *International Conference on Machine Learning*. PMLR, 2021, pp. 219–230.

[144] R. S. Sutton and A. G. Barto, *Reinforcement learning: An introduction*. MIT press, 2018.

[145] R. Bellman, "On the theory of dynamic programming," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 38, no. 8, p. 716, 1952.

[146] C. J. Watkins and P. Dayan, "Q-learning," *Machine learning*, vol. 8, no. 3-4, pp. 279–292, 1992.

[147] D. White, "Mean, variance, and probabilistic criteria in finite markov decision processes: A review," *Journal of Optimization Theory and Applications*, vol. 56, no. 1, pp. 1–29, 1988.

[148] T. Morimura, M. Sugiyama, H. Kashima, H. Hachiya, and T. Tanaka, "Parametric return density estimation for reinforcement learning," in *Proceedings of the Twenty-Sixth Conference on Uncertainty in Artificial Intelligence*, 2010, pp. 368–375.
[149] M. G. Azar, R. Munos, and B. Kappen, "On the sample complexity of reinforcement learning with a generative model," in *ICML*, 2012.
[150] M. G. Bellemare, W. Dabney, and R. Munos, "A distributional perspective on reinforcement learning," in *International Conference on Machine Learning*. PMLR, 2017, pp. 449–458.
[151] W. Dabney, M. Rowland, M. G. Bellemare, and R. Munos, "Distributional reinforcement learning with quantile regression," in *Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.
[152] M. Ghavamzadeh, S. Mannor, J. Pineau, and A. Tamar, "Bayesian reinforcement learning: A survey," *Foundations and Trends® in Machine Learning*, vol. 8, no. 5-6, pp. 359–483, 2015.
[153] A. M. Metelli, A. Likmeta, and M. Restelli, "Propagating uncertainty in reinforcement learning via wasserstein barycenters," in *33rd Conference on Neural Information Processing Systems, NeurIPS 2019*. Curran Associates, Inc., 2019, pp. 4335–4347.
[154] R. S. Sutton, D. McAllester, S. Singh, and Y. Mansour, "Policy gradient methods for reinforcement learning with function approximation," *Advances in neural information processing systems*, vol. 12, 1999.
[155] Y. Liu, P. Ramachandran, Q. Liu, and J. Peng, "Stein variational policy gradient," *UAI*, 2017.
[156] R. Zhang, C. Chen, C. Li, and L. Carin, "Policy optimization as wasserstein gradient flows," in *International Conference on Machine Learning*. PMLR, 2018, pp. 5737–5746.
[157] L. Ambrosio, N. Gigli, and G. Savaré, *Gradient flows: in metric spaces and in the space of probability measures*. Springer Science & Business Media, 2008.
[158] F. Santambrogio, "{Euclidean, metric, and Wasserstein} gradient flows: an overview," *Bulletin of Mathematical Sciences*, vol. 7, no. 1, pp. 87–154, 2017.
[159] R. Jordan, D. Kinderlehrer, and F. Otto, "The variational formulation of the fokker–planck equation," *SIAM journal on mathematical analysis*, vol. 29, no. 1, pp. 1–17, 1998.
[160] T. Haarnoja, H. Tang, P. Abbeel, and S. Levine, "Reinforcement learning with deep energy-based policies," in *International Conference on Machine Learning*. PMLR, 2017, pp. 1352–1361.
[161] R. M. Dudley, "The speed of mean glivenko-cantelli convergence," *The Annals of Mathematical Statistics*, vol. 40, no. 1, pp. 40–50, 1969.
[162] J. Weed and F. Bach, "Sharp asymptotic and finite-sample rates of convergence of empirical measures in wasserstein distance," *Bernoulli*, vol. 25, no. 4A, pp. 2620–2648, 2019.
[163] J. Niles-Weed and P. Rigollet, "Estimation of wasserstein distances in the spiked transport model," *arXiv preprint arXiv:1909.07513*, 2019.
[164] A. Korotin, L. Li, A. Genevay, J. M. Solomon, A. Filippov, and E. Burnaev, "Do neural optimal transport solvers work? a continuous wasserstein-2 benchmark." in *NeurIPS*, 2021.
[165] K. Nguyen and N. Ho, "Revisiting sliced wasserstein on images: From vectorization to convolution," *Advances in Neural Information Processing Systems*, vol. 35, pp. 17 788–17 801, 2022.
[166] C. Bonet, P. Berg, N. Courty, F. Septier, L. Drumetz, and M. T. Pham, "Spherical sliced-wasserstein," in *The Eleventh International Conference on Learning Representations*, 2022.
[167] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*. PMLR, 2017, pp. 1273–1282.
[168] C. Dwork, "Differential privacy: A survey of results," in *International conference on theory and applications of models of computation*. Springer, 2008, pp. 1–19.
[169] N. Lê Tien, A. Habrard, and M. Sebban, "Differentially private optimal transport: Application to domain adaptation." in *IJCAI*, 2019, pp. 2852–2858.
[170] T. Cao, A. Bie, A. Vahdat, S. Fidler, and K. Kreis, "Don't generate me: Training differentially private generative models with sinkhorn divergence," *Advances in Neural Information Processing Systems*, vol. 34, pp. 12 480–12 492, 2021.
[171] A. Rakotomamonjy, K. Nadjahi, and L. Ralaivola, "Federated wasserstein distance," *arXiv preprint arXiv:2310.01973*, 2023.
[172] F. E. Castellon, E. F. Montesuma, F. N. Mboula, A. Mayoue, A. Souloumiac, and C. Gouy-Pallier, "Federated dataset dictionary learning for multi-source domain adaptation," *arXiv preprint arXiv:2309.07670*, 2023.
[173] E. Fernandes Montesuma, F. N. Mboula, and A. Souloumiac, "Multi-source domain adaptation through dataset dictionary learning in wasserstein space," in *26th European Conference on Artificial Intelligence*, 2023.
[174] C. Cheng, B. Zhou, G. Ma, D. Wu, and Y. Yuan, "Wasserstein distance based deep adversarial transfer learning for intelligent fault diagnosis," *arXiv preprint arXiv:1903.06753*, 2019.
[175] E. F. Montesuma, M. Mulas, F. Corona, and F. M. N. Mboula, "Cross-domain fault diagnosis through optimal transport for a cstr process," *IFAC-PapersOnLine*, 2022.
[176] W. Dabney, Z. Kurth-Nelson, N. Uchida, C. K. Starkweather, D. Hassabis, R. Munos, and M. Botvinick, "A distributional code for value in dopamine-based reinforcement learning," *Nature*, vol. 577, no. 7792, pp. 671–675, 2020.

Eduardo Fernandes Montesuma was born in Fortaleza, Brazil, in 1996. He received a Bachelor degree in Computer Engineering from the Federal University of Ceará, and a M.S. degree in Electronics and Industrial Informatics from the National Institute of Applied Sciences of Rennes. He is currently working towards a PhD degree in Data Science at the University Paris-Saclay, and with the Department DIN of the CEA-LIST since 2021.

Fred Ngolè Mboula was born in Bandja city, Cameroon, in 1990. He received M.S. degree in Signal Processing from IMT Atlantique in 2013, and a PhD degree in astronomical images processing from Université Paris-Saclay in 2016. He joined the Department DIN of the CEA-LIST in 2017 as a researcher in ML and Signal Processing. His research interests include weakly supervised learning, transfer learning and information geometry. He is also teaching assistant at Université de Versailles Saint-Quentin in undergraduate mathematics.

Antoine Souloumiac was born in Bordeaux, France, in 1964. He received the M.S. and Ph.D. degrees in signal processing from Télécom Paris, France, in 1987 and 1993, respectively. From 1993 until 2001, he was Research Scientist with Schlumberger. He is currently with the Laboratory LS2D, CEA-LIST. His current research interests include statistical signal processing and ML, with emphasis on point processes, biomedical signal processing, and independent component analysis or blind source separation.